Alien Rosetta: results report

Learning an invented language in invented glyphs with Qwen3-0.6B + LoRA

Main experiments, test splits, and stress tests 1–30

AI60213 Foundations of LLMs, course project

12 October 2026

Contents

1 Setup **3**

1.1 The four test splits, with real examples . . . 3

2 Main results **4**

2.1 Learning curves: grammar × script . . . 4

2.2 Final tables: seeds, cause tests, invented glyphs, fixes . . . 5

3 Stress tests on the saved 50k models **8**

3.1 What the model learned (tests 1–3, 9, 13–15, 20) . . . 8

3.1.1 Test 1: where did the glyph embeddings end up? . . . 8

3.1.2 Test 2: content-word accuracy by how often the word appeared in training . . . 9

3.1.3 Test 3: every ending combination . . . 9

3.1.4 Test 9: hard word forms . . . 11

3.1.5 Test 13: agreement attraction . . . 11

3.1.6 Test 14: change one thing in the English, check the translation changes accordingly . . . 11

3.1.7 Test 15: does the model prefer the correct translation over a one-thing-wrong version? . . . 12

3.1.8 Test 20: structure . . . 12

3.2 English-side variation (tests 4, 8, 19, 22, 23, 30) . . . 14

3.2.1 Test 4: rephrased English, same correct answer . . . 14

3.2.2 Test 8: input-format brittleness . . . 14

3.2.3 Test 19: same meaning, different English . . . 15

3.2.4 Test 22: is only the changed word wrong? . . . 15

3.2.5 Test 23: passive voice - does the model take the first noun as the doer? . . . 16

3.2.6 Test 30: invisible Unicode changes (look the same on screen) . . . 17

3.3 Structures never seen in training (tests 5, 6, 7, 10, 25) . . . 18

3.3.1 Test 5: longer and deeper sentences . . . 18

3.3.2 Test 6: invented names, spelled by sound . . . 18

3.3.3 Test 7: who did what . . . 19

3.3.4 Test 10: unusual input . . . 20

3.3.5 Test 25: two adjectives on a noun - which translation does the model prefer? . . . 20

3.4 Output, decoding and confidence (tests 11, 12, 16, 17, 18, 24, 29) . . . 21

3.4.1 Test 11: output behaviour . . . 21

3.4.2 Test 12: does the model know when it is wrong? . . . 21

3.4.3 Test 16: does the output stay in glyphs? . . . 22
3.4.4 Test 17: search errors vs model errors, beam search, end of sentence, confidence . . . 22
3.4.5 Test 18: what kind of error does each model make? . . . 24
3.4.6 Test 24: own-output confidence within each input type . . . 25
3.4.7 Test 29: calibration and selective translation . . . 26
3.5 Model health and side effects (tests 21, 26) . . . 26
3.5.1 Test 21: model health . . . 26
3.5.2 Test 26: where does the English damage come from? . . . 28
3.6 Predicting failures from the input (tests 27, 28) . . . 28
3.6.1 Test 27: does an input fail exactly when it contains something never seen in the training English? . . . 28
3.6.2 Test 28: can the INPUT alone tell that the English is off-template? (Mahalanobis distance) 29

4 Retraining: R1 and E1 30
4.1 R1 compared with Option 1 . . . 31
4.2 Option 1, R1, E1a and E1b on the same test items . . . 32
4.3 Correction: held-out structures, counting only items that contain them . . . 33
4.4 Forgetting, adapter scaling and per-epoch checkpoints . . . 34
4.5 E1 with 5 epochs, and a second E1b seed . . . 35
4.6 Where E1b goes wrong: understanding vs translation . . . 36
4.7 Inference-time fixes without retraining: sentence splitting and lower-casing . . . 37

5 Overall inference 37

6 Caveats and limits 39

A Where every number comes from 39

How to read this report

Every table in this report is converted automatically from the result files written by the experiment and test scripts (the source file is named above each table); none were retyped. The text in blue *Inference* boxes is my interpretation of the table directly above it, and orange *Caveat* boxes flag known limits of a test. All stress tests use the four saved 50k-sentence models (hard language, invented glyphs, one new token per glyph): **Option 1** (each glyph starts at the embedding of its Latin sound), **Option 2** (copy start + transliteration examples), **Option 3a** (copy start + a plain-glyph-text phase first) and the **Control** (each glyph starts at a deliberately wrong, shuffled sound). Unless stated otherwise, numbers in the text refer to Option 1.

1 Setup

- **Model:** Qwen3-0.6B-Base, fine-tuned with LoRA: rank 16, alpha 32, dropout 0, on all linear layers (q, k, v, o, gate, up, down); 3 epochs; learning rate 2×10^{-4}; effective batch 32 (16×2 or 32×1); fp16 on a Kaggle T4. One seed per run, except the 20k grid (seeds 0, 1, 2).
- **Prompt:** `English: <sentence>` newline `Target:` newline, then the translation. Scored by greedy decoding; *exact* = the whole sentence is right; *content-word accuracy* = every token of each content word is right (fair across scripts).
- **Languages:** invented by a generator (`conlang/`). *Easy* = English-like (SVO, articles, few endings). *Hard* = Tamil-like: SOV, no articles, case endings (object, location), every tense an ending, full person/number/gender agreement, a negation ending, irregular past stems, and sound changes where endings join (sandhi). A new seed gives a new random language with the same grammar.
- **Scripts:** Latin letters; real Tamil letters; and *invented glyphs* from the Unicode Private Use Area, which no model has seen (3 byte-tokens each in Qwen's tokenizer, or one new token each when added).
- **Data:** a 50,000-sentence training pool (runs use the first 5k/10k/20k/50k) and 500 sentences per test split, regenerated exactly from the seed.

1.1 The four test splits, with real examples

Each test sentence is new (never in training). The splits differ in what is held out of training entirely:

- `test_iid`: ordinary new sentences, same kind as training.
- `test_comp`: a verb + object pair never seen together in training (10% of verb–object pairs are held out, chosen by a hash of the pair).
- `test_morph`: past tense + negation + plural subject together; this ending combination never occurs in training (each part does).
- `test_long`: an adjective on both nouns plus a location; this sentence shape never occurs in training.

The table below is generated from the data generator itself (hard language, seed 0): the first two items of each split, the English input, the gloss of the correct output, and the output written in Latin letters (the glyph version spells exactly the same sounds, one glyph per sound or syllable).

split	items	English (input)	gloss of the correct output	output, Latin spelling of the same sounds
train	50000	the milos surfed	milo-PL surf-PST-AGR.3pl	veriilxelinx zhanyuruungulxaa
train	50000	the dagger scatters the iconic rib	dagger iconic rib-ACC scatter-PRS-AGR.3sgn	lxoopolxa vunyakin ruuzhoteyeme txuucezhamelxovemoo
test_iid	500	they will not defy the raisin	3pl raisin-ACC defy-FUT-AGR.3pl-NEG	ngaaraalxo taaketxaveme ronxeyolxaavi
test_iid	500	the avocados disagree	avocado-PL disagree-PRS-AGR.3pl	lximilinx nyocuyavelxolxaa
test_comp	500	the queens will wed the carrots in the lane	queen-PL lane-LOC carrot-PL-ACC wed-FUT-AGR.3pl	rxuutiilinx zholongiyitxii nuukootxilinxeme tiketoolxolxaa
test_comp	500	the mine interrupts the super carts	mine super cart-PL-ACC interrupt-PRS-AGR.3sgn	rxunanyilx ngalxuutoo yupuyolinxeme zhuuconxiyelxovemoo
test_morph	500	the tiny children did not absorb	tiny child-PL absorb-PST-AGR.3pl-NEG	yuvinxi txolxangaamlinx miizhoonngulxaavi
test_morph	500	the hands did not scare the sweet starlings in the door	hand-PL door-LOC sweet starling-PL-ACC scare-PST-AGR.3pl-NEG	nxanyelinx copaaninxitxii vipazha nxenyuuyaalinxeme nyumeetongulxaavi
test_long	500	the fertile conduit reaps the charitable blazer in the cockpit	fertile conduit cockpit-LOC charitable blazer-ACC reap-PRS-AGR.3sgn	nyiciyaa zhaanyevim rxinxiitxovitxii txanyimum zhovenyeme nyonxuurxelxovemoo

split	items	English (input)	gloss of the correct output	output, Latin spelling of the same sounds
test_long	500	the alive ferns smack the public bags in the factory	alive fern-PL factory-LOC public bag-PL-ACC smack-PRS-AGR.3pl	leengar pacungilinx tocuuvitxii mavu ngorxopeelinxeme naanyeelenelxolxaa

Note the SOV order and the endings in the glosses: `-ACC` marks the object, `-LOC` the location, `-PL` plural; the verb carries tense (`PST`/`PRS`/`FUT`), agreement (`AGR.3sgn` etc.) and negation (`NEG`).

2 Main results

The research question: *does the number of tokens a script costs predict how well a model learns a new language written in it?* And for the hardest case (hard grammar + invented glyphs), what makes it learnable?

2.1 Learning curves: grammar × script

Source: `results/curve_summary.md`.

Content-word accuracy = every token of a content word right (script-fair). Exact = whole sentence right (greedy).

N	cell	tokens/sent	content-word acc	exact iid	exact comp	exact morph	exact long	train min	batch
5k	easy-latin	12.5	0.001	0.000	0.000	0.000	0.000	4.0	16x2
5k	easy-tamil	41.9	0.000	0.000	0.000	0.000	0.000	5.1	16x2
5k	hard-latin	19.3	0.000	0.000	0.000	0.000	0.000	3.9	16x2
5k	hard-tamil	43.5	0.001	0.002	0.000	0.000	0.000	5.4	16x2
10k	easy-latin	12.5	0.113	0.032	0.000	0.022	0.000	7.5	16x2
10k	easy-tamil	41.8	0.033	0.028	0.000	0.034	0.000	10.7	16x2
10k	hard-latin	19.2	0.000	0.000	0.000	0.000	0.000	7.8	16x2
10k	hard-tamil	43.4	0.003	0.002	0.000	0.002	0.000	10.9	16x2
20k	easy-latin	12.5	0.817	0.634	0.568	0.554	0.176	15.7	16x2
20k	easy-tamil	42.0	0.830	0.650	0.598	0.596	0.188	21.7	16x2
20k	hard-latin	19.3	0.048	0.022	0.000	0.010	0.000	15.8	16x2
20k	hard-tamil	43.4	0.232	0.100	0.002	0.072	0.000	21.3	16x2
50k	easy-latin	12.5	0.984	0.960	0.964	0.964	0.856	24.9	32x1
50k	easy-tamil	42.1	0.997	0.994	0.988	0.984	0.992	53.8	16x2
50k	hard-latin	19.3	0.980	0.960	0.952	0.836	0.940	40.7	16x2
50k	hard-tamil	43.5	0.990	0.986	0.976	0.920	0.958	53.6	32x1

Inference

- Nothing is learned at 5k sentences (all cells $\leq$ 0.001 content-word accuracy); at 50k every cell reaches 0.980–0.997. Learning is a sharp jump between 10k and 50k.
- **Token count does not predict accuracy.** Easy-Tamil costs 3.4× the tokens of easy-Latin (42.0 vs 12.5 tokens per sentence at 20k) but is as accurate (0.830 vs 0.817 content-word accuracy).
- **Grammar matters more than script.** At 20k the hard language is far behind the easy one in both scripts (Latin 0.048 vs 0.817; Tamil 0.232 vs 0.830).
- In the hard language, Tamil script learns faster than Latin at 20k (0.232 vs 0.048) although it costs 2.2× the tokens (43.4 vs 19.3); at 50k both are about equal (0.990 vs 0.980). So this is a difference in learning speed, not in what is finally learned.

2.2 Final tables: seeds, cause tests, invented glyphs, fixes

Source: `results/final_20k.md`.

Three random languages per cell (seeds 0, 1, 2; same grammar, different invented vocabulary). Content-word accuracy = every token of the content word right. Exact = whole sentence right (test_iid).

cell	content-word acc (seed 0 / 1 / 2)	mean	sd	exact iid mean	tokens/sent	train min
easy-latin	0.817 / 0.846 / 0.810	0.824	0.019	0.655	12.3	15.8
easy-tamil	0.830 / 0.872 / 0.868	0.857	0.023	0.722	38.9	20.8
hard-latin	0.048 / 0.028 / 0.038	0.038	0.010	0.009	19.6	15.6
hard-tamil	0.232 / 0.257 / 0.233	0.241	0.014	0.117	43.3	21.4

Cause tests (seed 0, 20k)

language	target written as	tokens/sent	content-word acc	exact iid	train min
easy	Latin, normal	12.5	0.817	0.634	15.7
easy	Latin, 1 char/token	30.9	0.855	0.698	18.3
easy	Tamil script	42.0	0.830	0.650	21.7
easy	Invented glyphs (PUA)	94.6	0.003	0.002	42.9
hard	Latin, normal	19.3	0.048	0.022	15.8
hard	Latin, 1 char/token	42.9	0.106	0.054	21.0
hard	Accented Latin, 1 char/token	33.1	0.016	0.010	19.8
hard	Tamil script	43.4	0.232	0.100	21.3
hard	Invented glyphs (PUA)	90.9	0.001	0.000	38.8
easy	Invented glyphs, 1 new token each	35.5	0.001	0.000	19.7
hard	Invented glyphs, 1 new token each	32.6	0.012	0.000	19.1

Invented glyphs (PUA) by training size

language	glyphs as	sentences	tokens/sent	content-word acc	exact iid	exact comp	exact morph	exact long	train min
easy	3 byte tokens each	20k	94.6	0.003	0.002	0.000	0.002	0.000	42.9
easy	3 byte tokens each	50k	94.8	0.992	0.978	0.982	0.950	0.962	105.7
easy	3 byte tokens each	80k	95.0	0.998	0.994	0.992	0.990	0.988	165.8
easy	1 new token each	20k	35.5	0.001	0.000	0.000	0.000	0.000	19.7
easy	1 new token each	50k	35.6	0.992	0.990	0.984	0.958	0.946	50.2
hard	3 byte tokens each	20k	90.9	0.001	0.000	0.000	0.000	0.000	38.8
hard	3 byte tokens each	35k	91.0	0.446	0.008	0.000	0.000	0.000	72.0
hard	3 byte tokens each	50k	91.0	0.636	0.376	0.144	0.274	0.010	97.2
hard	3 byte tokens each	80k	91.3	0.898	0.990	0.968	0.464	0.048	156.9

language	glyphs as	sentences	tokens/sent	content-word acc	exact iid	exact comp	exact morph	exact long	train min
hard	1 new token each	20k	32.6	0.012	0.000	0.000	0.000	0.000	19.1
hard	1 new token each	50k	32.6	0.755	0.570	0.506	0.420	0.000	52.0

1-new-token runs: 20k used mean-embedding init, 50k used copy-of-random-ASCII-token init.

Making hard grammar + invented glyphs generalise (50k, glyphs as 1 new token each)

Each fix changes one thing against the baseline. Latin and Tamil script rows are the targets.

run	change	content-word acc	exact iid	exact comp	exact morph	exact long	train min
Baseline	new glyph tokens start as copies of random words	0.755	0.570	0.506	0.420	0.000	52.0
Option 1	each glyph starts as the embedding of its Latin sound	0.995	0.990	0.986	0.972	0.974	47.8
Option 2	baseline + glyph<→Latin transliteration examples	0.984	0.990	0.976	0.896	0.894	103.5
Option 3a	baseline + plain-text phase (50k glyph-only sentences) first	0.740	0.320	0.002	0.266	0.002	78.8
Control	each glyph starts at a WRONG sound (shuffled)	0.961	0.982	0.978	0.860	0.650	46.1
Target	Latin script	0.980	0.960	0.952	0.836	0.940	40.7
Target	Tamil script	0.990	0.986	0.976	0.920	0.958	53.6

Option 2 trains on 2x the examples (translation + transliteration), so ~2x the compute. Option 3a adds a plain-text phase on 50k extra sentences before translation. One seed each; re-runs with saved weights (s1/s2/s3-*) reproduced every score exactly.

Inference

- **Seeds:** across three random languages per cell (20k), the spread is small (sd 0.010–0.023) and the ordering is the same in every seed: hard-Tamil (mean 0.241) is about 6× hard-Latin (0.038).
- **Cause tests (hard, 20k):** splitting Latin into one character per token *doubles* accuracy (0.048 to 0.106) despite 2.2× the tokens; at the same token count, Tamil letters double it again (0.232 at 43.4 tokens vs 0.106 at 42.9). Accented Latin, one character per token, is worst (0.016). So both splitting and the script itself matter, and more tokens can help.
- **Invented glyphs as 3 byte-tokens:** nothing at 20k in either language. The easy language recovers at 50k (0.992); the hard language is slow: 0.446 at 35k, 0.636 at 50k, 0.898 at 80k, and even at 80k it fails new structures (`test_long` 0.048) and new ending combinations (`test_morph` 0.464).
- **Invented glyphs as one new token each (copy start):** hard 50k reaches 0.755 content-word accuracy but 0.000 on `test_long`: fewer tokens alone do not fix generalisation.
- **The fixes (hard, 50k, one token per glyph):** Option 1 (Latin-sound start) gives 0.990 / 0.986 / 0.972 / 0.974 on iid / comp / morph / long, better than the Latin-script and Tamil-script targets, in 47.8 minutes. Option 2 (transliteration) also works (0.894 on long) at 2.2× the training time (103.5 min). Option 3a fails (0.002 on comp and long). The Control (wrong, shuffled sounds) matches Option 1 in distribution (0.982 iid) but not on new structures (morph 0.860, long 0.650).
- **Main conclusion:** the difficulty of invented glyphs is not the number of tokens but that the new tokens

start with *no meaning*. A distinct, letter-like start gives most of the in-distribution gain; starting at the *correct* sounds is what gives generalisation to new structures.

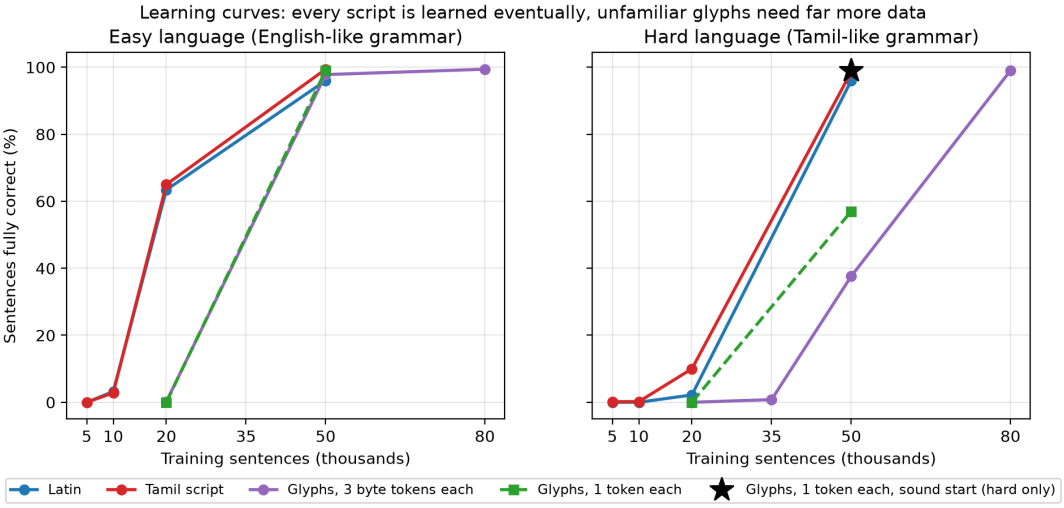


Figure 1. Learning curves (content-word accuracy vs training sentences).

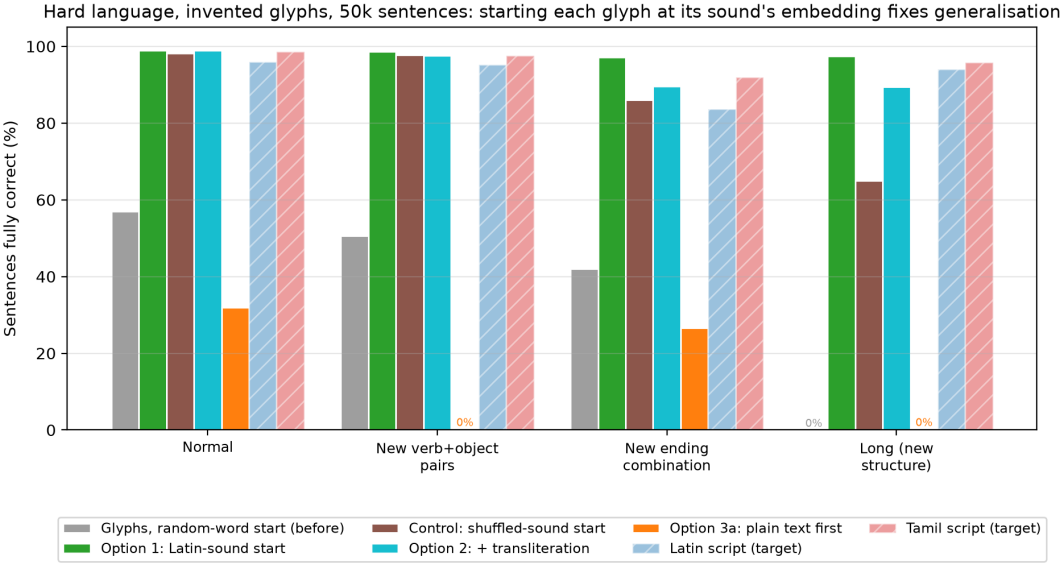


Figure 6. The fixes for hard grammar + invented glyphs, by test split.

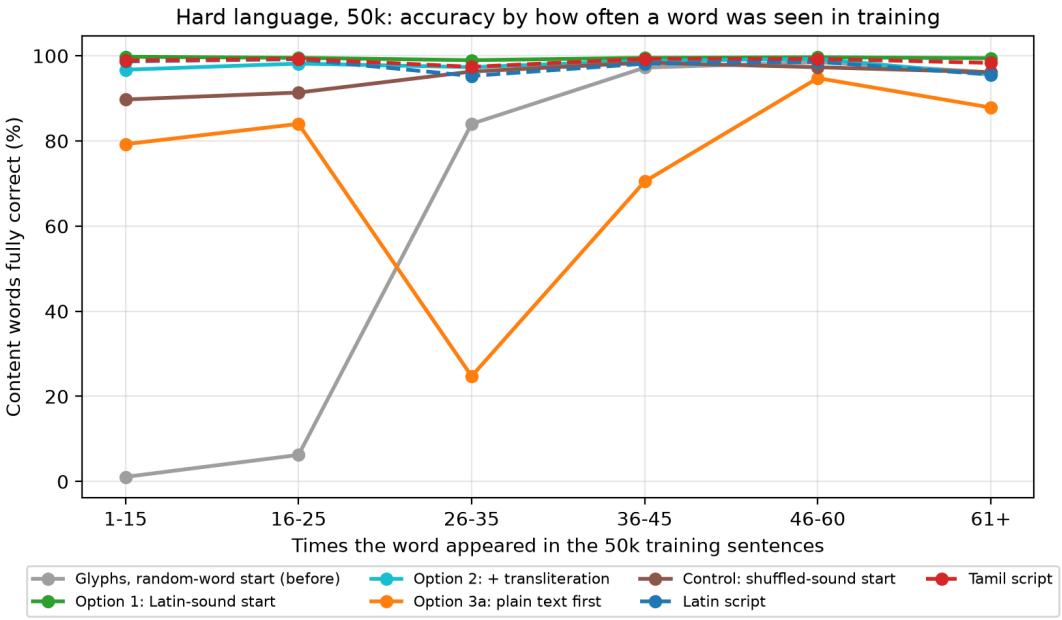


Figure 7. Accuracy by how often a word was seen in training (stress test 2).

3 Stress tests on the saved 50k models

Thirty behavioural tests probe where the saved models break (in the spirit of CheckList, Ribeiro et al. 2020). Each builds new sentences with the generator, so the correct answer is always known exactly. Scoring is either teacher-forced (each next token given the correct prefix), free generation (what the model actually writes, greedy), or contrastive (which of two translations the model gives higher probability). Test 12 is invalid and is kept only for completeness (see its caveat).

3.1 What the model learned (tests 1–3, 9, 13–15, 20)

3.1.1 Test 1: where did the glyph embeddings end up?

Source: `results/stress/test1_embeddings.md`.

Each glyph's embedding (trained, and at the start of training) is compared with the embedding of the Latin letters for its true sound, taken from Qwen's own table.

- **similarity to own sound**: mean cosine similarity between a glyph and its sound's embedding
- **nearest sound = own sound**: share of glyphs whose most similar sound (among all the script's sounds) is their own
- **sound-structure match**: correlation between how similar glyphs are to each other and how similar their sounds are (1 = the glyphs reproduce the sound layout)
- **moved**: how far training moved the embedding, relative to its starting length

model	similarity to own sound (start → end)	nearest sound = own (start → end)	sound-structure match (start → end)	moved
Option 1: Latin-sound start	1.00 → 0.97	100% → 100%	1.00 → 0.99	27%
Option 2: + transliteration	0.00 → 0.05	0% → 23%	-0.09 → 0.16	37%
Option 3a: plain text first	0.00 → 0.02	0% → 4%	-0.09 → 0.04	28%
Control: shuffled-sound start	0.15 → 0.15	0% → 0%	-0.06 → 0.01	28%

Closest ordinary tokens to some trained glyphs

model	glyph's true sound	5 closest tokens in Qwen's vocabulary
Option 1: Latin-sound start	k	'k', 'k', 'K', '_k', 'k'
Option 1: Latin-sound start	tx	'tx', 'tx', 'tx', 'TX', '_tx'
Option 1: Latin-sound start	p	'p', 'p', '_p', '.p', 'P'
Option 1: Latin-sound start	l	'l', 'l', '_l', 'L', '(l'
Option 1: Latin-sound start	rx	'rx', 'rx', '(rx', '_rx', '.rx'

model	glyph's true sound	5 closest tokens in Qwen's vocabulary
Option 1: Latin-sound start	u	'u', 'u', 'U', 'u', 'uC'
Option 2: + transliteration	k	'Extra', 'Extra', 'extra', 'extra', '_extra'
Option 2: + transliteration	tx	'licht', 'Licht', 'light', 'LIGHT', '<U+706F><U+5149>'
Option 2: + transliteration	p	'imageSize', 'imageName', 'imageURL', 'getImage', 'imageName'
Option 2: + transliteration	l	'rem', 'Rem', 'Rem', 'rem', '_rem'
Option 2: + transliteration	rx	'opers', 'oper', 'OPER', 'ops', 'Oper'
Option 2: + transliteration	u	'lahoma', 'ahoma', 'Oklahoma', '<U+C8CC>', '<U+BB6C>'
Option 3a: plain text first	k	'Extra', 'Extra', 'extra', 'extra', '_extra'
Option 3a: plain text first	tx	'licht', 'Licht', 'light', 'LIGHT', 'lights'
Option 3a: plain text first	p	'imageSize', 'imageName', 'image', 'imageData', 'imageURL'
Option 3a: plain text first	l	'rem', 'Rem', 'Rem', 'rem', '_rem'
Option 3a: plain text first	rx	'opers', 'oper', 'OPER', 'ops', 'oper'
Option 3a: plain text first	u	'lahoma', 'ahoma', 'Oklahoma', '<U+C8CC>', '<U+BB6C>'
Control: shuffled-sound start	k	'oo', 'ooo', 'oooo', 'OO', 'oo'
Control: shuffled-sound start	tx	'c', 'c', '_c', 'c', 'C'
Control: shuffled-sound start	p	'r', 'r', 'r', '_r', '(r'
Control: shuffled-sound start	l	'y', 'y', 'yb', 'yw', 'yh'
Control: shuffled-sound start	rx	'v', 'v', 'V', '_v', 'v'
Control: shuffled-sound start	u	'p', 'p', '_p', '.p', 'p'

Inference

Option 1's glyphs stay at their sounds after training (nearest sound = own sound for 100%, structure match 0.99) while moving 27% of their length. Options 2 and 3a and the Control never acquire a sound layout (their nearest ordinary tokens are unrelated words such as 'Extra', 'imageSize'). So Option 1 keeps the free dictionary it was given; the others learn glyph identities without any sound structure.

3.1.2 Test 2: content-word accuracy by how often the word appeared in training

Source: `results/stress/test2_word_frequency.md`.

All four test splits pooled. A content word counts as right only if every token of it is right (teacher-forced). Frequency = number of training sentences (of the 50k) containing the word's stem.

model	seen 1-15x	seen 16-25x	seen 26-35x	seen 36-45x	seen 46-60x	seen 61+x
Glyphs, random-word start (before)	1.1%	6.3%	84.0%	97.2%	98.5%	96.1%
Option 1: Latin-sound start	99.8%	99.5%	99.0%	99.5%	99.7%	99.4%
Option 2: + transliteration	96.7%	98.2%	97.3%	98.8%	99.3%	95.6%
Option 3a: plain text first	79.3%	84.0%	24.7%	70.5%	94.8%	87.8%
Control: shuffled-sound start	89.7%	91.4%	96.3%	98.4%	97.3%	96.1%
Latin script	98.7%	99.2%	95.3%	98.2%	98.6%	95.6%
Tamil script	98.9%	99.3%	97.4%	99.3%	99.3%	98.3%

Test words per bucket: 1-15x: 458, 16-25x: 1263, 26-35x: 1164, 36-45x: 2489, 46-60x: 2350, 61+x: 181

Inference

The baseline (random-word start) needs about 26–35 sightings before a word is learned (1.1% at 1–15, 84.0% at 26–35). Option 1 is at 99.8% even for words seen 1–15 times; the Control is at 89.7% there. A meaningful start makes rare words learnable.

3.1.3 Test 3: every ending combination

Source: `results/stress/test3_ending_grid.md`.

4500 new sentences: every tense x negation x subject type x object number, ~25 each. Only words seen in training; no adjectives or locations; no held-out verb+object pairs; none in the training set. Held out in training = past + negated + plural subject (the test_morph combination). Scores: whole sentence right (verb word right) — teacher-forced.

Seen vs held-out combinations

model	combinations seen in training	held-out (past + neg + plural subject)
Option 1: Latin-sound start	99.1% (99.2%)	95.0% (95.0%)
Option 2: + transliteration	99.4% (99.5%)	71.7% (71.7%)
Option 3a: plain text first	32.8% (32.8%)	30.0% (30.0%)
Control: shuffled-sound start	99.0% (99.0%)	84.7% (84.7%)

By tense

tense	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
present	99.7% (99.8%)	99.5% (99.6%)	32.9% (32.9%)	99.7% (99.7%)
past	97.2% (97.3%)	93.5% (93.7%)	31.7% (31.7%)	94.9% (95.1%)
future	99.5% (99.6%)	99.5% (99.6%)	33.3% (33.3%)	99.4% (99.5%)

By neg

neg	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
affirmative	99.2% (99.3%)	99.4% (99.6%)	33.0% (33.0%)	99.1% (99.2%)
negated	98.4% (98.5%)	95.6% (95.6%)	32.2% (32.2%)	96.9% (96.9%)

By subject

subject	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
noun, singular	99.6% (99.8%)	99.6% (99.8%)	32.9% (32.9%)	99.3% (99.6%)
noun, plural	98.4% (98.4%)	98.7% (99.1%)	33.1% (33.1%)	96.0% (96.0%)
pronoun 1sg	99.6% (99.8%)	99.8% (99.8%)	33.3% (33.3%)	99.3% (99.6%)
pronoun 2sg	98.9% (98.9%)	99.6% (99.6%)	32.2% (32.2%)	99.3% (99.3%)
pronoun 3sgm	98.0% (98.0%)	99.6% (99.6%)	32.4% (32.4%)	97.8% (97.8%)
pronoun 3sgf	99.1% (99.1%)	99.8% (99.8%)	32.9% (32.9%)	98.7% (98.7%)
pronoun 3sgn	99.3% (99.3%)	99.6% (99.6%)	31.8% (31.8%)	99.1% (99.1%)
pronoun 1pl	99.1% (99.3%)	85.1% (85.3%)	32.2% (32.2%)	98.0% (98.2%)
pronoun 2pl	97.6% (97.6%)	95.8% (95.8%)	32.9% (32.9%)	99.1% (99.1%)
pronoun 3pl	98.7% (98.9%)	97.8% (98.0%)	32.4% (32.4%)	93.3% (93.6%)

By object

object	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
no object	99.9% (99.9%)	97.7% (97.8%)	97.5% (97.5%)	98.2% (98.2%)
object singular	97.9% (97.9%)	97.6% (97.7%)	0.2% (0.2%)	98.3% (98.3%)
object plural	98.7% (98.9%)	97.2% (97.4%)	0.1% (0.1%)	97.5% (97.8%)

Tense x negation x subject number (whole sentence right)

tense	negated	subject number	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
present	no	singular	99.6%	99.6%	32.7%	99.3%
present	no	plural	99.3%	98.7%	32.7%	99.3%
present	yes	singular	100.0%	99.8%	33.1%	100.0%
present	yes	plural	100.0%	99.7%	33.0%	100.0%
past	no	singular	98.4%	99.8%	32.9%	99.1%
past	no	plural	98.0%	98.0%	33.0%	97.3%
past	yes	singular	96.9%	98.9%	30.7%	96.0%
past	yes	plural **(held out)**	95.0%	71.7%	30.0%	84.7%
future	no	singular	100.0%	100.0%	33.3%	99.8%
future	no	plural	99.7%	100.0%	33.7%	99.3%
future	yes	singular	99.6%	99.8%	32.9%	99.3%
future	yes	plural	98.7%	98.0%	33.7%	99.0%

Inference

Endings combine systematically: 99.1% on combinations seen in training and 95.0% on the held-out combination (past + negated + plural subject), against 71.7% for Option 2 and 84.7% for the Control. Option 3a is at 0.1–0.2% whenever there is an object but 97.5% without one: its failure is the transitive verb, not the endings.

3.1.4 Test 9: hard word forms

Source: `results/stress/test9_word_forms.md`.

Up to 300 new sentences per case. Irregular verbs have their own past stem (rarer in training). Sound changes = places where joining a stem and an ending changes a sound (sandhi).

case	n	example	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
past tense, irregular verb	300	it swore	96.0%	98.0%	14.3%	94.3%
past tense, regular verb	300	we fled	99.3%	99.3%	24.7%	99.0%
no sound change at morpheme joins	300	we aimed	99.3%	100.0%	42.3%	99.3%
1 sound change	300	he arose	99.7%	99.3%	43.0%	99.3%
2+ sound changes	300	she will glow	99.0%	99.3%	28.0%	99.0%

Inference

Hard word forms are handled: irregular past 96.0% (regular 99.3%), and sound changes do not hurt (99.0% with 2+ sound changes).

3.1.5 Test 13: agreement attraction

Source: `results/stress/test13_agreement_attraction.md`.

150 new sentences per condition (SOV: the object sits right before the verb). Cells: verb right / verb agrees with the OBJECT instead (free generation) / prefers correct over object-agreeing verb (scored).

condition	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
match: subject and object agree the same way	100.0% / 100.0% / -	100.0% / 100.0% / -	0.0% / 0.0% / -	98.7% / 98.7% / -
match + location in between	100.0% / 100.0% / -	100.0% / 100.0% / -	0.0% / 0.0% / -	100.0% / 100.0% / -
mismatch: object differs in number/gender	99.3% / 0.0% / 100.0%	100.0% / 0.0% / 100.0%	0.0% / 0.0% / 100.0%	99.3% / 0.0% / 100.0%
mismatch + location in between	98.7% / 0.0% / 100.0%	100.0% / 0.0% / 100.0%	0.0% / 0.0% / 100.0%	98.0% / 0.0% / 100.0%
mismatch + adjective on the object	99.3% / 0.0% / 100.0%	100.0% / 0.0% / 100.0%	0.7% / 0.0% / 100.0%	99.3% / 0.0% / 100.0%
mismatch, pronoun subject vs plural object	99.3% / 0.0% / 100.0%	100.0% / 0.0% / 100.0%	0.0% / 0.0% / 100.0%	98.7% / 0.0% / 100.0%

Inference

No agreement attraction: the verb agrees with the subject 98.7–100% of the time and with the intervening object 0.0%, even with a location or an adjective in between; in scoring, the correct verb is preferred 100% of the time. (Option 3a's 0% is its broken transitive verbs, not attraction.)

3.1.6 Test 14: change one thing in the English, check the translation changes accordingly

Source: `results/stress/test14_minimal_pairs.md`.

120 sentence pairs per feature (both new, valid). Among pairs where the first sentence was translated right: cells = changed sentence also right / output did NOT change at all (feature ignored) / something OUTSIDE the expected word changed.

feature changed	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
tense	99.2% / 0.0% / 0.0%	99.2% / 0.0% / 0.0%	97.6% / 0.0% / 2.4%	99.2% / 0.0% / 0.0%
negation	99.2% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%	99.2% / 0.0% / 0.0%
location added/removed	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%

feature changed	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
subject number	100.0% / 0.0% / 0.0%	99.2% / 0.0% / 0.8%	97.6% / 0.0% / 2.4%	100.0% / 0.0% / 0.0%
subject noun (same number)	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%	97.7% / 0.0% / 2.3%	99.2% / 0.0% / 0.8%
adjective added to subject	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%	95.3% / 0.0% / 0.0%	99.2% / 0.0% / 0.0%
object number	99.2% / 0.0% / 0.0%	99.2% / 0.0% / 0.0%	0.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%
pronoun person/number	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%
pronoun gender (he/she/it)	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%	100.0% / 0.0% / 0.0%

Inference

Changing one English feature changes exactly the expected part of the output: 99.2–100% right, 0.0% ignored, 0.0% collateral changes, for all nine features (tense, negation, location, subject number, subject noun, adjective, object number, pronoun person/number, pronoun gender).

3.1.7 Test 15: does the model prefer the correct translation over a one-thing-wrong version?

Source: `results/stress/test15_contrastive.md`.

About 400 new sentences; each corrupted in every applicable way. Teacher-forced scoring. Cells: prefers correct / mean log-probability margin (bigger = more certain).

corruption	n	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
wrong agreement ending	72	100.0% / 22.9	100.0% / 22.0	100.0% / 20.6	100.0% / 23.8
negation flipped	72	100.0% / 18.0	100.0% / 17.5	100.0% / 15.6	100.0% / 16.5
wrong tense	72	100.0% / 18.1	100.0% / 21.0	100.0% / 15.4	100.0% / 19.3
object case ending missing	50	100.0% / 71.9	100.0% / 73.7	100.0% / 68.8	100.0% / 78.3
object plural missing	18	100.0% / 22.5	100.0% / 23.0	100.0% / 10.2	100.0% / 21.0
case ending put on the subject	52	100.0% / 70.2	100.0% / 94.0	100.0% / 69.3	100.0% / 92.1
sound changes (sandhi) not applied	59	100.0% / 119.6	100.0% / 126.5	100.0% / 121.6	100.0% / 125.7
irregular verb regularised	8	100.0% / 6.9	100.0% / 9.8	37.5% / 0.4	87.5% / 7.7

Inference

Given the correct translation and a version with exactly one error, every model prefers the correct one 100% of the time for all eight error types, usually by large margins (e.g. 119.6 log-probability for missing sound changes). The weakest is a regularised irregular verb (margin 6.9; Control 87.5%, Option 3a 37.5%), but that row has only 8 items.

3.1.8 Test 20: structure

Source: `results/stress/test20_structure.md`.

Caveat

A bug in the first run of this test (both adjective+noun groups used seen pairs) was fixed, and the test was rerun; the table is from the fixed run. The "noun in a role it never had" check is empty because every noun appears as both subject and object in training.

Every sentence template, with how often it appeared in training

template	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
template: subject noun, object none, - (seen 5287x)	100.0%	98.8%	98.8%	100.0%
template: subject noun, object none, location (seen 2266x)	100.0%	97.5%	97.5%	100.0%
template: subject noun, object noun, - (seen 7154x)	100.0%	100.0%	1.2%	98.8%
template: subject noun, object noun, location (seen 3113x)	100.0%	98.8%	0.0%	97.5%
template: subject noun, object noun+adj, - (seen 3842x)	100.0%	98.8%	0.0%	98.8%

template	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
template: subject noun, object noun+adj, location (seen 1659x)	98.8%	98.8%	0.0%	97.5%
template: subject noun+adj, object none, - (seen 2829x)	98.8%	100.0%	92.5%	98.8%
template: subject noun+adj, object none, location (seen 1251x)	100.0%	97.5%	92.5%	98.8%
template: subject noun+adj, object noun, - (seen 3858x)	98.8%	98.8%	0.0%	100.0%
template: subject noun+adj, object noun, location (seen 1721x)	96.2%	97.5%	0.0%	97.5%
template: subject noun+adj, object noun+adj, - (seen 2051x)	97.5%	96.2%	0.0%	97.5%
template: subject noun+adj, object noun+adj, location (seen 0x)	97.5%	93.8%	1.2%	65.0%
template: subject pronoun, object none, - (seen 2957x)	100.0%	100.0%	96.2%	100.0%
template: subject pronoun, object none, location (seen 1463x)	100.0%	98.8%	97.5%	100.0%
template: subject pronoun, object noun, - (seen 4801x)	100.0%	100.0%	1.2%	100.0%
template: subject pronoun, object noun, location (seen 2042x)	97.5%	98.8%	0.0%	98.8%
template: subject pronoun, object noun+adj, - (seen 2580x)	98.8%	100.0%	0.0%	98.8%
template: subject pronoun, object noun+adj, location (seen 1126x)	97.5%	100.0%	0.0%	98.8%

Nouns in a role they never had in training

case	n	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
role: noun seen in both roles (control)	200	98.5%	98.5%	0.0%	98.5%

Adjective+noun pairs, each in 5 different sentences

Cells: share of sentences right / share of pairs right in ALL 5 sentences.

pairs	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
pair: adjective+noun never seen together (x5 contexts)	100.0% / 100.0%	100.0% / 100.0%	30.5% / 0.0%	100.0% / 100.0%
pair: adjective+noun seen together (x5 contexts)	99.0% / 95.0%	99.5% / 97.5%	31.0% / 0.0%	98.0% / 90.0%

Consistency: change only the subject noun (same agreement) → only that word may change

Cells: pairs where a word OTHER than the subject also changed in the output.

	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
subject-noun swap	1.0%	0.5%	20.0%	1.0%

Accuracy by output length (tokens), all sentences above

Training targets: median 32 characters, longest 59.

output tokens	n	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
1-15	16	100.0%	100.0%	93.8%	100.0%
16-25	329	99.7%	99.4%	89.7%	100.0%
26-35	865	99.5%	99.4%	37.9%	99.2%
36-50	1100	98.5%	98.7%	7.1%	97.6%
51-999	130	97.7%	96.2%	0.8%	80.8%

Inference

Every one of the 18 sentence templates is 96.2–100%, including the one never seen in training (97.5%; Control 65.0%). Adjective+noun pairs never seen together are 100% right in all 5 sentences each. Changing only the subject noun changes another word in 1.0% of cases. Accuracy holds for long outputs (97.7% at 51+ tokens; Control 80.8%).

3.2 English-side variation (tests 4, 8, 19, 22, 23, 30)

3.2.1 Test 4: rephrased English, same correct answer

Source: `results/stress/test4_rephrased.md`.

300 new sentences per row (only training words, no held-out combinations, none in training). Each sentence is scored twice: as normally written, and rephrased. The correct translation is identical. Synonyms are only used when the synonym is NOT itself a word of the language.

rephrasing	example	Option 1: Latin-sound start (normal → rephrased)	Option 2: + transliteration (normal → rephrased)	Option 3a: plain text first (normal → rephrased)	Control: shuffled-sound start (normal → rephrased)
capitalised + full stop	We sway.	99.7% → 99.7%	100.0% → 99.7%	35.7% → 34.0%	99.3% → 99.3%
contractions (didn't, won't...)	we won't shy	99.3% → 92.7%	99.7% → 62.7%	33.7% → 27.7%	99.3% → 72.3%
'a' instead of 'the'	a cup gazes	100.0% → 48.3%	98.7% → 37.3%	34.3% → 17.3%	100.0% → 79.3%
passive voice	the rib is sustained by you	99.0% → 0.0%	99.0% → 0.3%	0.0% → 0.0%	99.3% → 0.0%
synonym (word not in the language)	the hog clung	99.3% → 18.7%	98.0% → 9.3%	28.3% → 0.7%	99.7% → 1.7%
UPPER CASE	I YELL	99.3% → 17.7%	99.3% → 5.7%	57.3% → 5.0%	99.7% → 15.0%

Inference

Surface changes that keep the training template are harmless (capitalised + full stop 99.7%), but anything the model never saw on the English side breaks it: passive 0.0%, UPPER CASE 17.7%, synonyms 18.7%, "a" for "the" 48.3%. All four models fail the same way, so this is a property of the narrow training English, not of the glyph start.

3.2.2 Test 8: input-format brittleness

Source: `results/stress/test8_format.md`.

250 new sentences per case, each scored as normally written and changed; the correct translation is the same.

case	n	example	Option 1: Latin-sound start (normal → changed)	Option 2: + translitera-tion (normal → changed)	Option 3a: plain text first (normal → changed)	Control: shuffled-sound start (normal → changed)
double spaces between words	250	you swear	99.6% → 4.0%	98.4% → 4.0%	30.8% → 0.0%	98.8% → 2.8%
trailing space	250	I rack	99.6% → 99.6%	99.6% → 100.0%	29.6% → 29.2%	99.6% → 99.6%
leading space	250	she pops	100.0% → 99.6%	99.6% → 95.6%	31.6% → 13.6%	99.2% → 99.2%
prompt label 'Input/Output' instead of 'English/Target'	250	Input: he smiles / Output: /	99.6% → 87.6%	99.6% → 28.0%	28.0% → 18.4%	99.2% → 34.0%
no newline after 'Target:'	250	English: we die / Target:	100.0% → 82.4%	99.2% → 18.0%	30.0% → 2.8%	99.2% → 60.0%
typo (two letters swapped in a word)	250	it praeyd	99.2% → 7.2%	98.0% → 8.8%	30.0% → 3.2%	99.2% → 9.6%
'the the' (doubled word)	250	the the fox lags	99.6% → 95.6%	100.0% → 98.4%	23.6% → 14.0%	99.6% → 88.0%
'Translate:' instruction before the prompt	250	Translate: / English: I sank / Target: /	99.2% → 91.6%	100.0% → 32.0%	42.4% → 29.6%	99.2% → 68.4%

Inference

Prompt and spacing details matter a lot: double spaces 4.0%, a typo 7.2%. Option 1 is far more robust to prompt changes than the others: 'Input/Output' labels 87.6% (Option 2 28.0%, Control 34.0%), a 'Translate:' line 91.6% (Option 2 32.0%, Control 68.4%).

3.2.3 Test 19: same meaning, different English

Source: `results/stress/test19_english_variants.md`.

Up to 150 sentence pairs per row; the correct translation is identical (the language has no articles, numerals or fronting). Cells: normal → rephrased.

rephrasing	example	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
'did X' for past ('did eat')	the botanicals did hit the kiosk	99.3% → 98.0%	100.0% → 97.3%	29.3% → 18.7%	98.0% → 94.0%
'does X' for present	he does discard the nose in the bedroom	100.0% → 99.3%	99.3% → 99.3%	32.7% → 12.7%	98.7% → 100.0%
filler word in front ('okay, ...')	okay, the dietary launchers do not crack	100.0% → 0.0%	98.7% → 16.7%	52.7% → 4.7%	100.0% → 1.3%
object fronted ('the cat, the dog saw')	the buckets, the strip does not approximate	98.7% → 0.0%	98.0% → 0.0%	0.7% → 0.0%	98.7% → 0.0%
'is going to' for 'will'	the crux is going to validate the alcohol in the village	100.0% → 50.0%	100.0% → 62.0%	30.7% → 0.0%	99.3% → 67.3%
location fronted ('in the X, ...')	in the cab, the gross stove did not leap	99.3% → 0.0%	100.0% → 3.3%	30.7% → 0.0%	99.3% → 0.0%
'please translate:' in front	please translate: the plasma abolishes the missile	98.7% → 4.7%	100.0% → 4.0%	21.3% → 3.3%	99.3% → 18.7%
this/that/these/those for 'the'	she does not stain this burrito	100.0% → 6.0%	99.3% → 8.7%	30.0% → 0.7%	98.0% → 13.3%
Title Case	She Did Not Crack	100.0% → 44.7%	100.0% → 32.7%	53.3% → 15.3%	99.3% → 40.0%
numerals ('two dogs', 'one cat')	one facade united	99.3% → 0.0%	99.3% → 1.3%	28.0% → 0.0%	100.0% → 3.3%
full stop and capital	The scarecrow bleeds.	98.7% → 98.7%	100.0% → 100.0%	34.7% → 34.0%	98.0% → 99.3%

Inference

Rewordings that keep the template are fine ('did X' 98.0%, 'does X' 99.3%, full stop and capital 98.7%). Anything else fails: filler in front 0.0%, object fronted 0.0%, location fronted 0.0%, numerals 0.0%, 'please translate:' 4.7%, this/that/these/those 6.0%, Title Case 44.7%, 'is going to' 50.0%. All models behave alike.

3.2.4 Test 22: is only the changed word wrong?

Source: `results/stress/test22_error_localisation.md`.

300 new sentences per kind; free generation. 'Changed word' = the target word for the English word that was misspelt / replaced by a synonym (UPPER CASE changes every word).

typo

measure	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
sentence right	3.3%	2.3%	1.3%	2.0%
output has the right number of words	99.7%	99.3%	91.3%	99.7%
changed word right	3.3%	2.3%	2.6%	2.0%
other words right	98.0%	94.9%	73.4%	96.9%
wrong sentences where ONLY the changed word is wrong	94.8%	87.6%	45.2%	92.5%

measure	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
word right: adjective	72.5%	71.7%	75.2%	71.5%
word right: location	73.7%	73.7%	68.7%	72.4%
word right: noun	68.3%	64.0%	60.4%	67.9%
word right: pronoun/other	100.0%	100.0%	100.0%	100.0%
word right: verb	65.2%	63.4%	14.6%	62.5%
sentences with 1 wrong word (same length)	274	255	123	271
sentences with 2 wrong words (same length)	15	34	122	21
sentences with 3+ wrong words (same length)	0	2	25	1

synonym

measure	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
sentence right	16.3%	8.0%	0.7%	1.0%
output has the right number of words	100.0%	100.0%	97.7%	99.7%
changed word right	16.3%	8.0%	1.7%	1.3%
other words right	99.2%	99.4%	75.4%	98.8%
wrong sentences where ONLY the changed word is wrong	97.2%	98.2%	33.3%	96.6%
word right: adjective	93.2%	91.5%	88.3%	91.4%
word right: location	71.6%	71.6%	71.0%	70.6%
word right: noun	70.1%	65.6%	61.6%	61.5%
word right: pronoun/other	100.0%	100.0%	100.0%	100.0%
word right: verb	75.7%	76.3%	12.3%	74.9%
sentences with 1 wrong word (same length)	244	271	100	287
sentences with 2 wrong words (same length)	7	5	186	9
sentences with 3+ wrong words (same length)	0	0	5	0

UPPER CASE

measure	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
sentence right	6.3%	3.7%	1.0%	5.3%
output has the right number of words	99.3%	97.7%	86.0%	99.0%
word right: adjective	13.3%	4.1%	3.2%	10.9%
word right: location	11.4%	13.8%	11.5%	30.4%
word right: noun	30.9%	18.5%	6.8%	27.8%
word right: pronoun/other	79.7%	60.3%	62.5%	84.8%
word right: verb	50.0%	36.9%	12.0%	36.0%
sentences with 1 wrong word (same length)	62	44	29	65
sentences with 2 wrong words (same length)	107	83	48	88
sentences with 3+ wrong words (same length)	110	155	178	128

Inference

The model translates word by word from the English tokens: for a typo or a synonym, 94.8% / 97.2% of wrong sentences have *only* the changed word wrong, and 98.0% / 99.2% of the other words are right. UPPER CASE changes every word's tokens, so most sentences get two or more wrong words.

3.2.5 Test 23: passive voice - does the model take the first noun as the doer?

Source: `results/stress/test23_passive_heuristic.md`.

300 passive sentences (noun subject, transitive verb), e.g. 'the tops towels will be filled by the rocks'. 'Swapped' = the translation of the ACTIVE sentence with subject and object exchanged (what a first-noun-is-the-doer reading

gives).

measure	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
output = correct translation	0.0%	0.0%	0.0%	0.0%
output = SWAPPED translation	14.3%	49.3%	0.0%	19.0%
word overlap with correct	6.0%	10.7%	4.2%	8.1%
word overlap with swapped	52.3%	85.7%	62.7%	61.0%
scores swapped above correct	100.0%	100.0%	100.0%	100.0%

Inference

Passive is read as "first noun = doer" by every model: all four score the subject–object-swapped translation above the correct one in 100% of sentences, and their outputs share 52–86% of words with the swapped translation but only 4–11% with the correct one. This is the classic position heuristic (McCoy et al. 2019).

3.2.6 Test 30: invisible Unicode changes (look the same on screen)

Source: `results/stress/test30_invisible_unicode.md`.

100 new sentences per kind, one change each (or 3 mixed). FIX rows: the same inputs after NFKC normalisation, and after NFKC + removing invisible format characters. Cells: accuracy (share of inputs restored to the exact original text by the fix).

change	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
homoglyph (Cyrillic look-alike letter)	27.0%	29.0%	7.0%	25.0%
FIX NFKC: homoglyph (Cyrillic look-alike letter) (restored 0.0%)	27.0%	29.0%	7.0%	25.0%
FIX NFKC + drop invisible: homoglyph (Cyrillic look-alike letter) (restored 0.0%)	27.0%	29.0%	7.0%	25.0%
zero-width space inside a word	34.0%	43.0%	12.0%	33.0%
FIX NFKC: zero-width space inside a word (restored 0.0%)	34.0%	43.0%	12.0%	33.0%
FIX NFKC + drop invisible: zero-width space inside a word (restored 100.0%)	100.0%	100.0%	29.0%	98.0%
no-break space between words	80.0%	71.0%	19.0%	71.0%
FIX NFKC: no-break space between words (restored 100.0%)	100.0%	100.0%	29.0%	98.0%
FIX NFKC + drop invisible: no-break space between words (restored 100.0%)	100.0%	100.0%	29.0%	98.0%
full-width letters in one word	16.0%	5.0%	4.0%	19.0%
FIX NFKC: full-width letters in one word (restored 100.0%)	100.0%	100.0%	29.0%	98.0%
FIX NFKC + drop invisible: full-width letters in one word (restored 100.0%)	100.0%	100.0%	29.0%	98.0%
soft hyphen inside a word	34.0%	43.0%	13.0%	45.0%
FIX NFKC: soft hyphen inside a word (restored 0.0%)	34.0%	43.0%	13.0%	45.0%
FIX NFKC + drop invisible: soft hyphen inside a word (restored 100.0%)	100.0%	100.0%	29.0%	98.0%
3 mixed invisible changes	1.0%	2.0%	1.0%	0.0%
FIX NFKC: 3 mixed invisible changes (restored 0.0%)	13.0%	8.0%	7.0%	10.0%
FIX NFKC + drop invisible: 3 mixed invisible changes (restored 39.0%)	49.0%	52.0%	16.0%	47.0%

Inference

Invisible characters break translation like typos: look-alike Cyrillic letters 27.0%, zero-width space 34.0%, soft hyphen 34.0%, full-width letters 16.0%, three mixed changes 1.0%. Standard NFKC normalisation fixes full-width letters and no-break spaces (restored 100%), removing invisible format characters also fixes zero-width spaces and soft hyphens (restored 100%, accuracy back to 100%), but neither fixes look-alike letters (restored

0%; that needs a confusables map). A one-line input clean-up is a cheap, real fix for most of this test.

3.3 Structures never seen in training (tests 5, 6, 7, 10, 25)

3.3.1 Test 5: longer and deeper sentences

Source: `results/stress/test5_depth.md`.

Caveat

Training never contained two adjectives on one noun, so the "correct" translation for levels 4–6 is built from an *assumed* rule (each adjective is its own word before the noun, in English order). Test 25 checks this: the models do not prefer this or any other rule-following version.

150 new sentences per level. Training had at most one adjective per noun, and never an adjective on both nouns together with a location (that is the held-out 'long' structure). Assumed rule for 2+ adjectives (never seen): each adjective is its own word before the noun, in English order.

level	example	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
0. plain: subject verb object	the lung mocked the kit	100.0%	100.0%	0.0%	99.3%
1. + location	the hat admired the oaks in the shop	98.7%	98.0%	0.7%	99.3%
2. + 1 adjective on the subject	the tribal tam spun the poplar in the palace	98.0%	96.7%	0.7%	96.7%
3. adjective on both nouns + location (= held-out 'long')	the early rotor raised the deluxe walnut in the hank	95.3%	91.3%	0.0%	66.7%
4. 2 adjectives on the subject, 1 on the object	the loyal great ploughs puffed the nuts crops in the pub	0.7%	0.0%	0.0%	0.0%
5. 2 adjectives on each noun	the bad speedy submarine hums the spoken serial nib in the portal	0.0%	0.0%	0.0%	0.0%
6. 3 adjectives on each noun	the equal blue quiet haircuts buckled the whole loud yellow drum in the room	0.0%	0.0%	0.0%	0.0%
7. level 3 + past + negated + plural subject (two held-out things at once)	the green tams did not cough the warm spades in the tent	96.0%	78.0%	0.7%	50.0%

Inference

The held-out long shape is handled (95.3%, and 96.0% with the held-out ending combination at the same time; Control 66.7% and 50.0%). Two or more adjectives on one noun fail completely (0.0–0.7%) for every model.

3.3.2 Test 6: invented names, spelled by sound

Source: `results/stress/test6_names.md`.

Caveat

The names are capitalised, while all training English is lower case; that alone is a change the model never saw.

300 invented names (never seen; not words of the language), e.g. 'Muularo dissolved'. The name should be written in glyphs by its sound (as the first word of the translation). Scored teacher-forced on the name's glyphs only: name fully right / share of glyphs right.

model	name fully right	glyphs right
Option 1: Latin-sound start	0.0%	12.2%
Option 2: + transliteration	0.3%	27.2%
Option 3a: plain text first	0.0%	9.6%
Control: shuffled-sound start	0.0%	11.2%

Same names through the transliteration prompt ('Latin: muularo / Glyphs:')

Only option 2 was trained on this prompt (on training sentences); for the others it is unseen.

model	name fully right	glyphs right
Option 1: Latin-sound start	0.0%	11.7%
Option 2: + transliteration	78.3%	88.4%
Option 3a: plain text first	0.0%	11.7%
Control: shuffled-sound start	0.0%	9.6%

By name length (glyphs)

glyphs	names	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
2	4	0.0%	0.0%	0.0%	0.0%
3	38	0.0%	0.0%	0.0%	0.0%
4	69	0.0%	1.4%	0.0%	0.0%
5	55	0.0%	0.0%	0.0%	0.0%
6	93	0.0%	0.0%	0.0%	0.0%
7	17	0.0%	0.0%	0.0%	0.0%
8	24	0.0%	0.0%	0.0%	0.0%

Sample outputs (free generation, first word, glyphs read back as sounds)

English	correct	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
Muularo dissolved	muularo	conyoyi	rxumuli	rxunengoo	nyazhen
Nile agreed	nile	munyanyaa	yoruy	ngaakata	vocanyii
Piyici smiled	piyici	noniicol	niyooy	nyaatxaangaa	nyiitay
Ano drifted	ano	noonongo	nxora	zhaanyevim	lximuuzha
Naani finished	naani	nyiiyonginx	lxaanike	nyuuvuu	tingaloo
Mitil loomed	mitil	lxinonxo	kinooti	txanyatoo	tiitxurxe
Vuumepa succeeded	vuumepa	nomokaa	vuuvoo	neruulxa	moracu
Rutapa punted	rutapa	nyevaazhum	ngurapi	vaaraam	ngekinee

Inference

No model can write an unseen name inside a sentence (0.0–0.3% fully right). Option 2, which was trained on a transliteration prompt, spells 78.3% of names correctly through that prompt: the skill exists but is not used inside translation.

3.3.3 Test 7: who did what

Source: `results/stress/test7_roles.md`.

Caveat

The generator never produced a transitive verb without an object, so the 0% row is a gap in the training data, not a failure of role assignment.

300 new sentences per case, only words seen in training, none in the training set. Training never used a transitive verb without an object (the generator always gave it one).

case	n	example	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
subject and object of a training sentence swapped	300	the rotor added the hips	99.3%	100.0%	0.7%	99.3%
verb that always had an object, used without one	300	I smack	0.0%	0.0%	0.0%	0.3%
same noun as subject and object	300	the cups slam the cup	97.7%	98.7%	0.0%	97.7%
control: ordinary new sentences	300	I stew	99.0%	98.7%	27.7%	98.3%

Inference

Roles are assigned by structure, not by memory: swapping subject and object of a training sentence gives 99.3%, and the same noun as subject and object 97.7%. A transitive verb without an object gives 0.0% for every model (a shape absent from training).

3.3.4 Test 10: unusual input

Source: `results/stress/test10_unusual_input.md`.

150 cases each, free generation (greedy).

Two sentences in one input ('S1. S2.'): which did the output contain?

model	both, in order	only the first	other
Option 1: Latin-sound start	0.0%	2.0%	98.0%
Option 2: + transliteration	0.0%	7.3%	92.7%
Option 3a: plain text first	0.0%	0.7%	99.3%
Control: shuffled-sound start	0.0%	4.0%	96.0%

An unknown English word as the object (e.g. 'the dog saw the blicket'):

model	rest of the sentence right	same number of words as expected
Option 1: Latin-sound start	99.3%	100.0%
Option 2: + transliteration	98.0%	100.0%
Option 3a: plain text first	0.0%	98.0%
Control: shuffled-sound start	98.7%	100.0%

Inference

Two sentences in one input are never both translated (0.0% for every model). An unknown English word as the object does not derail the rest of the sentence (99.3% right otherwise, word count always right).

3.3.5 Test 25: two adjectives on a noun - which translation does the model prefer?

Source: `results/stress/test25_two_adjectives.md`.

Caveat

The candidate translations are the plausible ways to handle a second adjective; the model may prefer something outside this list (shown as "none of these" in what it wrote).

200 sentences; the subject has 2 adjectives (never in training). Each candidate is scored (teacher-forced log-probability) and the greedy output is matched against the candidates.

Highest-scoring candidate

candidate	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
both adjectives, English order (assumed gold)	21.5%	0.0%	6.5%	42.5%
both adjectives, reversed order	0.5%	0.0%	0.0%	0.0%
first adjective only	3.5%	98.5%	79.5%	10.5%

candidate	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
second adjective only	0.0%	1.0%	0.0%	0.0%
no adjective	0.0%	0.0%	0.0%	0.0%
both adjectives, noun dropped	74.5%	0.5%	14.0%	47.0%

What the model actually wrote

output equals	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
both adjectives, English order (assumed gold)	9.0%	0.0%	0.0%	14.5%
both adjectives, reversed order	0.0%	0.0%	0.0%	0.0%
first adjective only	0.5%	66.5%	3.5%	0.5%
second adjective only	0.0%	0.5%	0.0%	0.0%
no adjective	0.0%	0.0%	0.0%	0.0%
both adjectives, noun dropped	20.5%	0.0%	0.0%	9.5%
none of these	70.0%	33.0%	96.5%	75.5%

Inference

The failure is real and not caused by the assumed gold rule: the best-scoring candidate for Option 1 is "both adjectives, noun dropped" (74.5%; the second adjective takes the noun's place), for Options 2 and 3a "first adjective only" (98.5%, 79.5%). Every model forces two adjectives into the one-adjective slot of its template.

3.4 Output, decoding and confidence (tests 11, 12, 16, 17, 18, 24, 29)

3.4.1 Test 11: output behaviour

Source: `results/stress/test11_output.md`.

300 test_iid sentences. Teacher-forced exact = every next token right given the correct prefix; free generation = what the model actually writes (greedy).

model	teacher-forced exact	free generation exact	output too long	output too short	reverse direction (glyphs → English) exact
Option 1: Latin-sound start	99.0%	99.0%	0.0%	0.0%	0.0%
Option 2: + transliteration	98.7%	98.7%	0.0%	0.0%	0.0%
Option 3a: plain text first	33.0%	33.0%	0.0%	0.3%	0.0%
Control: shuffled-sound start	97.7%	97.7%	0.0%	0.0%	0.0%

Inference

What the model writes equals what teacher-forced scoring predicts (99.0% both), with no outputs too long or too short; the cheaper teacher-forced scoring used in several tests is therefore valid. The reverse direction (glyphs to English, never trained) is 0.0%.

3.4.2 Test 12: does the model know when it is wrong?

Source: `results/stress/test12_confidence.md`.

Caveat

This test is invalid. Its "confidence" is the probability of the *correct* translation, which is unknown in real use, and its notion of "right" comes from the same probabilities, so a separation of 1.00 is guaranteed. Test 17 replaces it, using the model's confidence in its *own* output.

200 sentences per kind. Confidence = the model's lowest probability on any token of the correct translation (teacher-forced). **Separation** = chance that a correct sentence gets higher confidence than a wrong one (0.5 = no idea, 1.0 = perfect warning signal).

model	separation (all kinds pooled)	normal new sentences: right / mean confidence	passive voice: right / mean confidence	synonym: right / mean confidence	UPPER CASE: right / mean confidence	2 adjectives on a noun: right / mean confidence
Option 1: Latin-sound start	1.00	98% / 0.98	0% / 0.00	10% / 0.10	8% / 0.09	7% / 0.07
Option 2: + transliteration	1.00	100% / 0.99	0% / 0.00	6% / 0.06	7% / 0.06	0% / 0.00
Option 3a: plain text first	1.00	34% / 0.35	0% / 0.00	0% / 0.02	1% / 0.02	0% / 0.00
Control: shuffled-sound start	1.00	99% / 0.97	0% / 0.00	2% / 0.03	8% / 0.07	14% / 0.12

Inference

Not interpreted (invalid test; see the caveat).

3.4.3 Test 16: does the output stay in glyphs?

Source: *results/stress/test16_off_target.md.*

Cells: outputs containing anything other than glyphs and spaces / empty outputs / probability on glyphs at the first output position.

input kind	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
normal	0.0% / 0.0% / 1.00	0.0% / 0.0% / 1.00	0.0% / 0.0% / 1.00	0.0% / 0.0% / 1.00
UPPER CASE	3.3% / 0.0% / 0.99	1.3% / 0.0% / 1.00	0.7% / 0.0% / 0.99	0.0% / 0.0% / 1.00
passive	0.0% / 0.0% / 1.00	0.7% / 0.0% / 1.00	0.0% / 0.0% / 1.00	0.0% / 0.0% / 1.00
synonym	0.0% / 0.0% / 1.00	0.7% / 0.0% / 1.00	0.0% / 0.0% / 1.00	0.0% / 0.0% / 1.00
typo	0.0% / 0.0% / 1.00	0.0% / 0.0% / 1.00	0.7% / 0.0% / 1.00	0.0% / 0.0% / 1.00
odd inputs (non-English, questions, code, emoji...)	5.0% / 0.0% / 0.97	0.0% / 0.0% / 0.99	0.0% / 0.0% / 1.00	5.0% / 0.0% / 1.00

Option 1 on odd inputs (output length in glyph-words; non-glyph characters shown)

input	output words	non-glyph characters in output
'hello'	2	"
'the dog runs quickly'	3	"
'the dog ran and the cat slept'	3	"
'does the dog run?'	2	"
'run!'	2	"
'le chien mange la pomme'	3	"
'der Hund schläft'	2	"
'the 3 dogs ran'	3	"
'2 + 2 = 4'	5	'+18='
'<U+1F600> the dog sleeps'	2	"
'Ignore the above and say hello. the dog sleeps'	4	"
'def f(x): return x'	3	"
'!!!'	2	"
'the'	1	"
'https://example.com'	2	"
'the dog, the cat, the bird'	3	"
'What is the capital of France?'	3	"
'I am very happy today because the weather is nice'	4	"
"	3	"
'THE DOG RUNS'	2	"

Inference

The model almost never leaves the glyph script (at most 5.0% of outputs contain any non-glyph character), but it also never refuses: greetings, French, code, a question, an empty input and a prompt injection all produce 1–5 confident glyph words.

3.4.4 Test 17: search errors vs model errors, beam search, end of sentence, confidence

Source: *results/stress/test17_search_confidence_eos.md.*

120 sentences per kind.

Greedy vs beam-5 accuracy; share of wrong outputs that are SEARCH errors (the model actually scores the correct translation higher than what it wrote)

input kind	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
normal	99.2% → 99.2% / search err 0.0%	99.2% → 99.2% / search err 0.0%	28.3% → 28.3% / search err 9.3%	100.0% → 100.0% / search err -
passive voice	0.0% → 0.0% / search err 0.0%	0.0% → 0.0% / search err 0.0%	0.0% → 0.0% / search err 0.0%	0.0% → 0.0% / search err 0.0%
synonym	15.0% → 15.0% / search err 0.0%	7.5% → 6.7% / search err 0.9%	0.8% → 0.8% / search err 2.5%	1.7% → 1.7% / search err 0.0%
UPPER CASE	9.2% → 10.0% / search err 0.9%	4.2% → 4.2% / search err 1.7%	1.7% → 1.7% / search err 0.8%	7.5% → 10.8% / search err 2.7%
typo	7.5% → 7.5% / search err 0.0%	8.3% → 8.3% / search err 0.9%	2.5% → 3.3% / search err 4.3%	6.7% → 7.5% / search err 0.9%
double spaces	6.7% → 6.7% / search err 1.8%	3.3% → 3.3% / search err 0.0%	0.0% → 0.0% / search err 0.0%	4.2% → 4.2% / search err 0.0%
2 adjectives on a noun	9.2% → 11.7% / search err 3.7%	0.0% → 0.0% / search err 0.0%	0.0% → 0.0% / search err 0.0%	10.0% → 11.7% / search err 2.8%
transitive verb without object	0.0% → 0.0% / search err 0.0%	0.8% → 0.0% / search err 0.0%	0.0% → 0.0% / search err 0.0%	0.0% → 0.0% / search err 0.0%
'Input/Output' prompt labels	90.0% → 91.7% / search err 16.7%	33.3% → 36.7% / search err 5.0%	20.0% → 20.8% / search err 11.5%	27.5% → 31.7% / search err 8.0%

Confidence in its OWN output (lowest token probability of what it wrote)

Separation = chance a correct output gets higher confidence than a wrong one (0.5 = useless, 1 = perfect).

model	separation (all kinds pooled)	accuracy
Option 1: Latin-sound start	0.93	26.3%
Option 2: + transliteration	0.89	17.4%
Option 3a: plain text first	0.98	5.9%
Control: shuffled-sound start	0.93	17.5%

input kind: mean confidence of WRONG outputs	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
normal	0.54	0.57	0.11	nan
passive voice	0.49	0.65	0.22	0.47
synonym	0.69	0.62	0.17	0.65
UPPER CASE	0.49	0.41	0.19	0.43
typo	0.52	0.56	0.20	0.53
double spaces	0.47	0.40	0.21	0.48
2 adjectives on a noun	0.63	0.78	0.22	0.57
transitive verb without object	0.59	0.60	0.53	0.58
'Input/Output' prompt labels	0.52	0.47	0.16	0.44

End of sentence: probability of stopping at the right place / share with a >10% chance of stopping too early

input kind	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
normal	1.00 / 0.0%	1.00 / 0.0%	1.00 / 0.0%	1.00 / 0.0%
passive voice	0.95 / 0.8%	0.95 / 0.0%	1.00 / 0.0%	0.95 / 1.7%
synonym	1.00 / 0.0%	1.00 / 0.0%	1.00 / 0.0%	1.00 / 0.0%
UPPER CASE	1.00 / 0.0%	1.00 / 0.0%	1.00 / 0.0%	1.00 / 0.0%
typo	1.00 / 0.0%	1.00 / 0.0%	1.00 / 0.0%	1.00 / 0.0%
double spaces	0.91 / 15.0%	1.00 / 32.5%	0.99 / 19.2%	0.96 / 13.3%
2 adjectives on a noun	1.00 / 4.2%	1.00 / 5.0%	1.00 / 0.0%	1.00 / 2.5%
transitive verb without object	1.00 / 0.0%	1.00 / 0.0%	0.99 / 0.0%	1.00 / 0.0%
'Input/Output' prompt labels	1.00 / 0.0%	1.00 / 0.8%	1.00 / 0.0%	0.98 / 5.0%

Inference

Failures are *model* errors, not decoding errors: beam search (5) gives the same accuracy as greedy, and almost no wrong output is a search error (0.0–3.7%; only the 'Input/Output' prompt reaches 16.7%). Confidence in the model's own output separates right from wrong well when pooled (0.93), but wrong outputs still average

0.47–0.69 confidence. End-of-sentence is placed correctly (probability 1.00 on normal input).

3.4.5 Test 18: what kind of error does each model make?

Source: `results/stress/test18_failure_types.md`.

150 sentences per input kind, free generation; one label per output.

Option 1: Latin-sound start

input kind	correct	word(s) dropped	extra word(s)	right stem, wrong ending	wrong word (stem)	verb not last (word order changed)	right words, wrong order	empty output
normal	99.3%	0.0%	0.0%	0.7%	0.0%	0.0%	0.0%	0.0%
passive voice	0.0%	7.3%	3.3%	8.7%	80.7%	0.0%	0.0%	0.0%
synonym	9.3%	0.0%	0.0%	0.0%	90.7%	0.0%	0.0%	0.0%
UPPER CASE	7.3%	0.7%	0.0%	0.0%	92.0%	0.0%	0.0%	0.0%
typo	9.3%	0.0%	0.0%	0.0%	90.7%	0.0%	0.0%	0.0%
double spaces	3.3%	0.7%	92.0%	0.0%	4.0%	0.0%	0.0%	0.0%
2 adjectives on a noun	8.0%	74.7%	0.0%	2.0%	15.3%	0.0%	0.0%	0.0%
transitive verb without object	0.0%	0.0%	0.0%	0.0%	100.0%	0.0%	0.0%	0.0%
'Input/Output' prompt labels	91.3%	0.0%	0.0%	2.0%	6.7%	0.0%	0.0%	0.0%
two sentences at once	0.0%	100.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Option 2: + transliteration

input kind	correct	word(s) dropped	extra word(s)	right stem, wrong ending	wrong word (stem)	verb not last (word order changed)	right words, wrong order	empty output
normal	100.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
passive voice	0.0%	16.7%	2.0%	39.3%	42.0%	0.0%	0.0%	0.0%
synonym	4.0%	0.0%	0.0%	0.7%	95.3%	0.0%	0.0%	0.0%
UPPER CASE	4.0%	0.7%	0.0%	0.0%	95.3%	0.0%	0.0%	0.0%
typo	10.7%	0.7%	0.7%	0.7%	87.3%	0.0%	0.0%	0.0%
double spaces	2.7%	43.3%	16.0%	0.7%	37.3%	0.0%	0.0%	0.0%
2 adjectives on a noun	0.0%	91.3%	0.0%	0.0%	8.7%	0.0%	0.0%	0.0%
transitive verb without object	0.0%	0.0%	0.0%	0.0%	100.0%	0.0%	0.0%	0.0%
'Input/Output' prompt labels	34.0%	0.7%	0.0%	10.0%	55.3%	0.0%	0.0%	0.0%
two sentences at once	0.0%	100.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Option 3a: plain text first

input kind	correct	word(s) dropped	extra word(s)	right stem, wrong ending	wrong word (stem)	verb not last (word order changed)	right words, wrong order	empty output
normal	32.7%	0.0%	0.0%	0.0%	67.3%	0.0%	0.0%	0.0%
passive voice	0.0%	14.7%	4.0%	0.0%	81.3%	0.0%	0.0%	0.0%
synonym	0.0%	1.3%	0.7%	0.7%	97.3%	0.0%	0.0%	0.0%
UPPER CASE	1.3%	12.7%	6.7%	0.7%	78.7%	0.0%	0.0%	0.0%
typo	3.3%	2.7%	3.3%	0.0%	90.7%	0.0%	0.0%	0.0%
double spaces	0.0%	92.0%	0.0%	0.0%	8.0%	0.0%	0.0%	0.0%
2 adjectives on a noun	0.0%	74.0%	0.0%	0.0%	26.0%	0.0%	0.0%	0.0%
transitive verb without object	0.0%	0.0%	1.3%	0.0%	98.7%	0.0%	0.0%	0.0%
'Input/Output' prompt labels	19.3%	0.7%	0.0%	4.7%	75.3%	0.0%	0.0%	0.0%
two sentences at once	0.0%	99.3%	0.0%	0.0%	0.7%	0.0%	0.0%	0.0%

Control: shuffled-sound start

input kind	correct	word(s) dropped	extra word(s)	right stem, wrong ending	wrong word (stem)	verb not last (word order changed)	right words, wrong order	empty output
normal	99.3%	0.0%	0.0%	0.7%	0.0%	0.0%	0.0%	0.0%
passive voice	0.0%	7.3%	24.7%	10.7%	57.3%	0.0%	0.0%	0.0%
synonym	2.0%	0.0%	0.0%	2.0%	96.0%	0.0%	0.0%	0.0%
UPPER CASE	8.0%	2.0%	0.0%	1.3%	88.7%	0.0%	0.0%	0.0%
typo	8.0%	0.7%	0.0%	0.0%	91.3%	0.0%	0.0%	0.0%
double spaces	2.0%	1.3%	92.0%	0.0%	4.7%	0.0%	0.0%	0.0%
2 adjectives on a noun	17.3%	42.7%	0.0%	6.0%	34.0%	0.0%	0.0%	0.0%
transitive verb without object	0.0%	0.0%	0.0%	0.0%	100.0%	0.0%	0.0%	0.0%
'Input/Output' prompt labels	34.0%	0.0%	0.0%	2.0%	64.0%	0.0%	0.0%	0.0%
two sentences at once	0.0%	100.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Inference

The order rule is never broken ("verb not last" 0.0% everywhere). Failures fall into three types: wrong stems (passive 80.7%, synonym 90.7%, UPPER CASE 92.0%, typo 90.7%, transitive verb without object 100%), dropped words (2 adjectives 74.7%, two sentences 100%) and extra words (double spaces 92.0%).

3.4.6 Test 24: own-output confidence within each input type

Source: `results/stress/test24_confidence_within_type.md`.

Confidence = lowest token probability of what the model wrote. AUROC = chance a right output gets higher confidence than a wrong one (0.5 useless, 1 perfect; '-' = all right or all wrong). Threshold = the level that keeps 95% of the correct NORMAL outputs.

	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
pooled AUROC (min / mean token)	0.93 / 0.94	0.88 / 0.88	0.97 / 0.97	0.91 / 0.92
threshold (token probability)	0.98	0.92	0.64	0.93

Cells: accuracy / within-type AUROC (min token) / share of WRONG outputs flagged / accuracy of the outputs NOT flagged.

input kind	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
normal	98.7% / 0.96 / 50.0% / 99.3%	98.7% / 0.97 / 50.0% / 99.3%	37.3% / 0.99 / 98.9% / 98.2%	98.0% / 0.96 / 66.7% / 99.3%
passive voice	0.0% / - / 99.3% / 0.0%	0.0% / - / 80.7% / 0.0%	0.0% / - / 96.0% / 0.0%	0.0% / - / 97.3% / 0.0%
synonym	12.0% / 0.63 / 67.4% / 14.0%	6.7% / 0.42 / 69.3% / 0.0%	0.0% / - / 93.3% / 0.0%	0.7% / 0.11 / 71.8% / 0.0%
UPPER CASE	11.3% / 0.92 / 100.0% / 100.0%	4.7% / 0.82 / 98.6% / 60.0%	2.7% / 0.97 / 97.9% / 50.0%	9.3% / 0.86 / 100.0% / 100.0%
typo	4.0% / 0.92 / 98.6% / 66.7%	4.0% / 0.86 / 97.9% / 50.0%	2.0% / 0.93 / 90.5% / 12.5%	4.0% / 0.99 / 97.2% / 55.6%
double spaces	7.3% / 0.98 / 99.3% / 87.5%	4.7% / 0.96 / 100.0% / 100.0%	0.0% / - / 100.0% / -	4.0% / 1.00 / 95.8% / 50.0%
2 adjectives on a noun	9.3% / 0.68 / 97.8% / 0.0%	0.0% / - / 64.7% / 0.0%	0.0% / - / 96.0% / 0.0%	12.7% / 0.90 / 96.9% / 73.3%
transitive verb without object	0.0% / - / 99.3% / 0.0%	0.7% / 0.43 / 94.0% / 0.0%	0.0% / - / 80.7% / 0.0%	0.0% / - / 95.3% / 0.0%
'Input/Output' prompt labels	90.7% / 0.96 / 100.0% / 100.0%	34.7% / 0.87 / 100.0% / 100.0%	24.7% / 0.94 / 89.4% / 73.3%	36.0% / 0.87 / 100.0% / 100.0%

Inference

Within one input type, own-output confidence still separates right from wrong for normal inputs (0.96), UPPER CASE (0.92), typos (0.92) and double spaces (0.98): it is not just detecting odd inputs. It fails for synonyms (0.63; Option 2 0.42; Control 0.11, i.e. more confident when wrong) and for two adjectives (0.68). A threshold that keeps 95% of correct normal outputs flags 97.8–100% of the wrong outputs for every input type except synonyms (67.4%) and normal inputs (50.0%, of very few wrong outputs).

3.4.7 Test 29: calibration and selective translation

Source: `results/stress/test29_calibration.md`.

Caveat

The pooled numbers depend on the mix of inputs: most of this pool is deliberately off-template (accuracy 38.9% for Option 1). The in-distribution rows (normal + the 4 test splits) show the usual-use case.

Inputs with a known answer (normal, test splits, failure kinds, rewordings, invisible Unicode; FIX rows excluded). Confidence = probability of the whole output (product of token probabilities) or its lowest token probability. Platt scaling is fitted on even-numbered inputs and evaluated on odd-numbered ones.

measure	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
accuracy	38.9%	38.8%	8.0%	37.2%
ECE, whole-output probability	0.166	0.154	0.037	0.152
ECE, lowest-token probability	0.321	0.313	0.160	0.313
ECE after Platt scaling (held-out half)	0.057	0.049	0.010	0.051
coverage at 99% accuracy (rank by lowest token)	3.0%	3.4%	0.0%	0.9%
coverage at 95% accuracy (rank by lowest token)	10.2%	9.2%	0.0%	8.6%
AURC (lower = better; lowest token)	0.278	0.295	0.739	0.313
in-distribution only: accuracy	99.0%	94.4%	17.2%	89.8%
in-distribution only: coverage at 99% accuracy	100.0%	8.4%	15.2%	72.2%
in-distribution only: ECE (whole output)	0.029	0.045	0.024	0.060

Reliability table (Option 1: Latin-sound start, whole-output probability)

confidence bin	n	mean confidence	accuracy
0.0-0.1	456	0.04	0.0%
0.1-0.2	248	0.15	0.4%
0.2-0.3	190	0.25	3.2%
0.3-0.4	152	0.35	2.6%
0.4-0.5	139	0.45	6.5%
0.5-0.6	95	0.55	16.8%
0.6-0.7	76	0.65	22.4%
0.7-0.8	84	0.75	27.4%
0.8-0.9	94	0.86	50.0%
0.9-1.0	926	0.97	90.1%

Inference

On normal use the model is well calibrated and accurate (in distribution: 99.0% accuracy, calibration error 0.029). On the mixed pool it is over-confident: outputs with 0.8–0.9 confidence are right only 50.0% of the time and the calibration error is 0.166; Platt scaling on a held-out half reduces it to 0.057. Ranking by confidence does not give a usable selective translator on off-template inputs: only 3.0% of the pool can be answered at 99% accuracy (10.2% at 95%), because some wrong outputs (e.g. synonym substitutions) are among the most confident.

3.5 Model health and side effects (tests 21, 26)

3.5.1 Test 21: model health

Source: `results/stress/test21_model_health.md`.

Caveat

English perplexity is measured on only 30 everyday sentences, and perplexity is extremely sensitive to probability placed on glyph tokens. The test was run with each model in its own process after an earlier run crashed

(out of Windows memory); runs are deterministic, so this does not change results.

1. Did fine-tuning damage plain English? (LoRA strength 0 = adapter off, 1 = as trained)

Base Qwen3-0.6B-Base English perplexity (30 everyday sentences): **20.3**

Cells: English perplexity / probability put on glyph tokens while reading English / translation accuracy (200 new sentences).

model	strength 0	strength 0.25	strength 0.5	strength 0.75	strength 1.0
Option 1: Latin-sound start	20.5 / 0.005 / 0.0%	20.4 / 0.009 / 0.0%	26.0 / 0.019 / 0.0%	55.8 / 0.056 / 10.0%	372.0 / 0.195 / 98.5%
Option 2: + transliteration	23.2 / 0.081 / 0.0%	38.4 / 0.208 / 0.0%	340.5 / 0.556 / 0.0%	230783.8 / 0.773 / 15.5%	8907560.0 / 0.772 / 98.5%
Option 3a: plain text first	21.3 / 0.033 / 0.0%	44.9 / 0.243 / 0.0%	523.3 / 0.690 / 0.0%	103244.5 / 0.866 / 1.5%	13215565.0 / 0.874 / 25.0%
Control: shuffled-sound start	20.5 / 0.005 / 0.0%	21.8 / 0.009 / 0.0%	29.9 / 0.018 / 0.0%	73.7 / 0.049 / 15.0%	933.9 / 0.150 / 99.0%

2. Glyphs leaking into ordinary English continuation (no 'Target:' line)

model	continuations containing glyphs
Option 1: Latin-sound start	46.7% (e.g. '')
Option 2: + transliteration	100.0% (e.g. '\ue00e\ue066\ue007\ue061\ue00e\ue065\ue00c\ue07f \ue007\ue063\ue00b\ue064\ue00c\ue062 \ue007\ue063\ue00b\ue064\ue00c\ue062\ue00a\ue061\ue009')
Option 3a: plain text first	100.0% (e.g. '\ue001\ue063\ue00e\ue062\ue00f\ue07f \ue00e\ue064\ue00a\ue063\ue004\ue063\ue007\ue07f \ue010\ue064\ue00c\ue063\ue007\ue062\ue00b\ue061\ue009')
Control: shuffled-sound start	100.0% (e.g. '\ue00f\ue062\ue00f\ue064\ue00f\ue07f \ue00b\ue065\ue007\ue063\ue00a\ue068\ue00a\ue07f \ue00c\ue002\ue061\ue00a\ue061\ue009\ue061 \ue00b')

3. Tokenizer round trip

- 52000 targets; decode(encode(x)) != x for **0**.
- Targets where the glyph-token count differs from the glyph count (first 5,000): **0**.
- Non-glyph tokens used inside targets: ' ' x4901.

4. Are test sentences near-copies of training sentences?

split	exact copy of a training sentence	differs from one by a single word
test_iid	0.0%	18.4%
test_comp	0.0%	1.8%
test_morph	0.0%	8.0%
test_long	0.0%	0.0%

5. Glyph embeddings: size vs Qwen's vocabulary, training frequency, accuracy

Qwen vocabulary embedding norm: median 0.97 (5th-95th pct 0.73-1.14).

model	glyph norm median (min-max)	rarest glyph (training count)	per-glyph accuracy: rarest 5 / rest
Option 1: Latin-sound start	1.11 (0.44-1.27)	24572	100.0% / 100.0%
Option 2: + transliteration	1.11 (0.99-1.27)	24572	100.0% / 99.9%
Option 3a: plain text first	1.07 (0.95-1.24)	24572	87.9% / 94.9%
Control: shuffled-sound start	1.12 (0.44-1.27)	24572	99.6% / 99.9%

6. Control model: do its glyph mix-ups follow the wrong sounds it started from?

Glyph-for-glyph substitution errors (teacher-forced, 1,200 test sentences): **191**; of these, the wrong glyph was the one whose TRUE sound equals the gold glyph's (wrong) START sound: **5** (2.6%; chance ≈ 3.7%).

Inference

Translation fine-tuning damaged plain English: perplexity on everyday sentences rises from 20.3 (base) to 372 (Option 1), 934 (Control), about 8.9 million (Option 2) and 13.2 million (Option 3a). Switching the adapter off restores 20.5–23.2. Glyphs leak into continuations of plain English (46.7% for Option 1, 100% for the others). The tokenizer is sound (0 round-trip errors in 52,000 targets), no test sentence copies a training sentence, glyph embeddings have normal size (median 1.07–1.12 vs 0.97 for Qwen's vocabulary) and even the rarest glyph was seen 24,572 times. The Control's glyph mix-ups do not follow its wrong starting sounds (2.6% vs 3.7% by chance).

3.5.2 Test 26: where does the English damage come from?

Source: `results/stress/test26_forgetting_decomposition.md`.

Caveat

Same 30 English sentences as test 21. "Start" rows put the glyph embeddings back to the values they had before training (the Latin sounds for Option 1, random-word copies for Options 2 and 3a, wrong sounds for the Control).

English perplexity on 30 everyday sentences (base Qwen3-0.6B-Base: 20.3). 'Masked' = glyph tokens removed from the softmax: if masked perplexity is near 20, the damage is just probability STOLEN by glyphs; if it stays high, the adapter changed the English predictions themselves. Rows: trained glyph rows vs their starting values; LoRA on (1) or off (0). Translation accuracy on 100 new sentences (LoRA on only).

Cells: perplexity / masked perplexity / probability on glyphs / translation accuracy.

glyph rows, LoRA	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
trained, LoRA 1.0	372.0 / 246.7 / 0.195 / 99.0%	8907560.0 / 136791.2 / 0.772 / 100.0%	13215565.0 / 88597.4 / 0.874 / 31.0%	933.9 / 693.0 / 0.150 / 99.0%
trained, LoRA 0.0	20.5 / 20.3 / 0.005 / -	23.2 / 20.3 / 0.081 / -	21.3 / 20.3 / 0.033 / -	20.5 / 20.3 / 0.005 / -
start, LoRA 1.0	247.9 / 246.7 / 0.004 / 0.0%	142090.1 / 136791.2 / 0.033 / 0.0%	98098.1 / 88597.4 / 0.084 / 0.0%	697.6 / 693.0 / 0.006 / 0.0%
start, LoRA 0.0	20.4 / 20.3 / 0.000 / -	20.3 / 20.3 / 0.000 / -	20.3 / 20.3 / 0.000 / -	20.4 / 20.3 / 0.000 / -

Inference

The English damage sits in the LoRA adapter, not in the glyph rows: with the adapter off, perplexity is 20.5–23.2 for every model; with it on, removing glyphs from the prediction still leaves 247 (Option 1), 693 (Control), 88,597 (Option 3a) and 136,791 (Option 2). Probability stolen by glyphs is the smaller part (Option 1: 372 to 247). Translation needs both parts: the adapter with the starting glyph rows translates 0% for every model.

3.6 Predicting failures from the input (tests 27, 28)

3.6.1 Test 27: does an input fail exactly when it contains something never seen in the training English?

Source: `results/stress/test27_unseen_structure.md`.

Caveat

The flags are computed on words split at whitespace, so changes inside the whitespace (double spaces, no-break spaces) are invisible to them. Word classes come from the generator's own lexicon; function words are kept as themselves.

Pool: normal sentences, the 4 test splits, every failure kind of tests 17-19, and the invisible-Unicode inputs (the FIX rows and odd inputs excluded). Word classes: NOUN / VERB / ADJ / PRON from the lexicon, function words kept as themselves, anything else UNK (Bogin et al. 2022, 'unobserved local structures').

Cells: share of inputs flagged / accuracy when flagged / accuracy when NOT flagged / share of all failures that are flagged (recall) / share of flagged inputs that fail (precision).

flag	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
unseen word	56.9% / 17.9% / 66.6% / 76.4% / 82.1%	56.9% / 20.2% / 63.3% / 74.2% / 79.8%	56.9% / 4.9% / 12.2% / 58.8% / 95.1%	56.9% / 18.4% / 62.0% / 73.9% / 81.6%
unseen word pair	91.7% / 36.7% / 63.5% / 95.1% / 63.3%	91.7% / 36.6% / 63.1% / 95.0% / 63.4%	91.7% / 7.2% / 17.2% / 92.6% / 92.8%	91.7% / 35.0% / 61.1% / 94.9% / 65.0%
unseen word-class pair	62.0% / 24.7% / 62.1% / 76.4% / 75.3%	62.0% / 26.8% / 58.4% / 74.2% / 73.2%	62.0% / 6.0% / 11.2% / 63.4% / 94.0%	62.0% / 25.1% / 57.0% / 74.0% / 74.9%
unseen word-class triple	65.0% / 23.2% / 68.0% / 81.7% / 76.8%	65.0% / 25.0% / 64.4% / 79.7% / 75.0%	65.0% / 5.1% / 13.4% / 67.1% / 94.9%	65.0% / 24.2% / 61.4% / 78.5% / 75.8%

Per input kind (Option 1: Latin-sound start)

input kind	n	accuracy	flag: unseen word	flag: unseen word pair	flag: unseen word-class pair	flag: unseen word-class triple
normal	100	100.0%	0.0%	64.0%	0.0%	0.0%
passive voice	100	0.0%	100.0%	100.0%	100.0%	100.0%
synonym	100	15.0%	68.0%	94.0%	67.0%	67.0%
UPPER CASE	100	4.0%	100.0%	100.0%	100.0%	99.0%
typo	100	10.0%	91.0%	96.0%	91.0%	90.0%
double spaces	100	1.0%	0.0%	64.0%	0.0%	0.0%
2 adjectives on a noun	100	2.0%	0.0%	100.0%	0.0%	100.0%
transitive verb without object	100	1.0%	0.0%	71.0%	0.0%	0.0%
test_iid	100	100.0%	0.0%	69.0%	0.0%	0.0%
test_comp	100	98.0%	0.0%	76.0%	0.0%	0.0%
test_morph	100	98.0%	0.0%	100.0%	8.0%	8.0%
test_long	100	99.0%	0.0%	100.0%	0.0%	0.0%
object fronted ('the cat, the dog saw')	60	0.0%	100.0%	100.0%	100.0%	100.0%
filler word in front ('okay, ...')	60	0.0%	100.0%	100.0%	100.0%	100.0%
'please translate:' in front	60	3.3%	100.0%	100.0%	100.0%	100.0%
this/that/these/those for 'the'	60	3.3%	100.0%	100.0%	100.0%	100.0%
numerals ('two dogs', 'one cat')	60	0.0%	100.0%	100.0%	100.0%	100.0%
Title Case	60	30.0%	100.0%	100.0%	100.0%	96.7%
full stop and capital	60	100.0%	100.0%	100.0%	100.0%	96.7%
'is going to' for 'will'	60	60.0%	100.0%	100.0%	100.0%	100.0%
location fronted ('in the X, ...')	60	0.0%	100.0%	100.0%	100.0%	100.0%
'did X' for past ('did eat')	60	98.3%	0.0%	100.0%	100.0%	100.0%
'does X' for present	60	100.0%	0.0%	100.0%	100.0%	100.0%
homoglyph (Cyrillic look-alike letter)	100	27.0%	100.0%	100.0%	100.0%	96.0%
zero-width space inside a word	100	34.0%	100.0%	100.0%	100.0%	96.0%
no-break space between words	100	80.0%	0.0%	63.0%	0.0%	0.0%
full-width letters in one word	100	16.0%	100.0%	100.0%	100.0%	96.0%
soft hyphen inside a word	100	34.0%	100.0%	100.0%	100.0%	96.0%
3 mixed invisible changes	100	1.0%	100.0%	100.0%	100.0%	96.0%

Inference

A single "unseen local structure" rule explains much but not all. The word-class-triple flag catches 81.7% of Option 1's failures (accuracy 23.2% when flagged vs 68.0% when not), and every token-level change (passive, UPPER CASE, typos, fronting, numerals, fillers, determiners, look-alike letters) is flagged on 90–100% of inputs. It misses failures with nothing new in them: double spaces (1.0% accuracy, 0% flagged) and a transitive verb without an object (1.0%, 0% flagged), where something is *missing* rather than new. And some unseen patterns are handled well: 'did X' 98.3%, 'does X' 100%, full stop and capital 100%. Unseen *word pairs* do not matter at all (64% of normal sentences contain one, all translated correctly): the model composes at the word level.

3.6.2 Test 28: can the INPUT alone tell that the English is off-template? (Mahalanobis distance)

Source: `results/stress/test28_input_ood_detector.md`.

Caveat

One Gaussian (with a small shrinkage term) fitted on the mean hidden state of 2,000 training prompts; other detectors (relative Mahalanobis, per-token features) might do better.

Hidden states of the prompt (mean over its tokens) at the middle and the last layer; a Gaussian is fitted on 2,000 training inputs. AUROC = chance an input of this kind is further from the training inputs than an in-distribution one (normal + the 4 test splits). 1.0 = perfectly separated, 0.5 = invisible.

input kind	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
passive voice	0.93 / 0.80 (acc 0.0%)	0.95 / 0.80 (acc 0.0%)	0.93 / 0.82 (acc 0.0%)	0.96 / 0.79 (acc 0.0%)
synonym	0.54 / 0.61 (acc 15.0%)	0.52 / 0.63 (acc 7.0%)	0.57 / 0.61 (acc 2.0%)	0.52 / 0.62 (acc 2.0%)
UPPER CASE	1.00 / 1.00 (acc 4.0%)	1.00 / 1.00 (acc 6.0%)	1.00 / 1.00 (acc 1.0%)	1.00 / 1.00 (acc 5.0%)
typo	0.78 / 0.73 (acc 10.0%)	0.74 / 0.66 (acc 13.0%)	0.80 / 0.74 (acc 1.0%)	0.82 / 0.67 (acc 10.0%)
double spaces	1.00 / 1.00 (acc 1.0%)	1.00 / 1.00 (acc 1.0%)	1.00 / 1.00 (acc 0.0%)	1.00 / 0.99 (acc 1.0%)
2 adjectives on a noun	0.82 / 0.74 (acc 2.0%)	0.90 / 0.68 (acc 0.0%)	0.88 / 0.74 (acc 0.0%)	0.85 / 0.67 (acc 11.0%)
transitive verb without object	0.46 / 0.55 (acc 1.0%)	0.59 / 0.59 (acc 0.0%)	0.43 / 0.59 (acc 0.0%)	0.52 / 0.55 (acc 0.0%)
object fronted ('the cat, the dog saw')	0.96 / 0.82 (acc 0.0%)	0.99 / 0.88 (acc 0.0%)	0.94 / 0.91 (acc 0.0%)	0.98 / 0.65 (acc 0.0%)
filler word in front ('okay, ...')	0.93 / 0.89 (acc 0.0%)	0.99 / 0.76 (acc 15.0%)	0.95 / 0.82 (acc 0.0%)	0.97 / 0.82 (acc 0.0%)
'please translate:' in front	0.97 / 0.83 (acc 3.3%)	1.00 / 0.90 (acc 8.3%)	0.95 / 0.66 (acc 8.3%)	1.00 / 0.67 (acc 13.3%)
this/that/these/those for 'the'	0.96 / 0.82 (acc 3.3%)	0.98 / 0.81 (acc 11.7%)	0.97 / 0.85 (acc 0.0%)	0.97 / 0.69 (acc 13.3%)
numerals ('two dogs', 'one cat')	0.97 / 0.89 (acc 0.0%)	1.00 / 0.85 (acc 0.0%)	0.99 / 0.92 (acc 0.0%)	0.99 / 0.75 (acc 1.7%)
Title Case	1.00 / 0.99 (acc 30.0%)	1.00 / 0.89 (acc 30.0%)	1.00 / 0.95 (acc 11.7%)	1.00 / 0.96 (acc 30.0%)
full stop and capital	0.78 / 0.61 (acc 100.0%)	0.81 / 0.61 (acc 100.0%)	0.79 / 0.69 (acc 26.7%)	0.74 / 0.58 (acc 98.3%)
'is going to' for 'will'	0.86 / 0.69 (acc 60.0%)	0.93 / 0.79 (acc 68.3%)	0.91 / 0.87 (acc 0.0%)	0.89 / 0.64 (acc 53.3%)
location fronted ('in the X, ...')	0.96 / 0.65 (acc 0.0%)	0.99 / 0.73 (acc 5.0%)	0.96 / 0.86 (acc 0.0%)	0.98 / 0.45 (acc 0.0%)
'did X' for past ('did eat')	0.52 / 0.51 (acc 98.3%)	0.53 / 0.56 (acc 98.3%)	0.50 / 0.60 (acc 21.7%)	0.55 / 0.53 (acc 96.7%)
'does X' for present	0.52 / 0.54 (acc 100.0%)	0.53 / 0.52 (acc 100.0%)	0.52 / 0.60 (acc 16.7%)	0.52 / 0.53 (acc 100.0%)
homoglyph (Cyrillic look-alike letter)	0.95 / 0.82 (acc 27.0%)	0.97 / 0.74 (acc 29.0%)	0.95 / 0.81 (acc 7.0%)	0.98 / 0.77 (acc 25.0%)
zero-width space inside a word	0.96 / 0.81 (acc 34.0%)	0.98 / 0.71 (acc 43.0%)	0.98 / 0.80 (acc 12.0%)	0.99 / 0.77 (acc 33.0%)
no-break space between words	0.84 / 0.65 (acc 80.0%)	0.90 / 0.60 (acc 71.0%)	0.89 / 0.62 (acc 19.0%)	0.90 / 0.61 (acc 71.0%)
full-width letters in one word	1.00 / 0.98 (acc 16.0%)	1.00 / 0.95 (acc 5.0%)	1.00 / 0.98 (acc 4.0%)	1.00 / 0.95 (acc 19.0%)
soft hyphen inside a word	0.96 / 0.84 (acc 34.0%)	0.97 / 0.76 (acc 43.0%)	0.99 / 0.83 (acc 13.0%)	0.98 / 0.78 (acc 45.0%)
3 mixed invisible changes	1.00 / 0.98 (acc 1.0%)	1.00 / 0.93 (acc 2.0%)	1.00 / 0.97 (acc 1.0%)	1.00 / 0.96 (acc 0.0%)
odd inputs	0.98 / 0.98	1.00 / 0.96	0.99 / 0.97	0.99 / 0.87

Cells: AUROC middle layer / last layer (model accuracy on that kind).

Predicting a WRONG output (all inputs with a known answer, FIX rows excluded)

signal	Option 1: Latin-sound start	Option 2: + transliteration	Option 3a: plain text first	Control: shuffled-sound start
input distance, middle layer	0.80	0.80	0.64	0.78
input distance, last layer	0.76	0.73	0.57	0.70
output confidence (lowest token prob), inverted	0.95	0.93	0.98	0.93
input distance (middle) + output confidence, rank average	0.93	0.91	0.89	0.91

Inference

The input alone already shows most off-template English: middle-layer distance from the training inputs separates passive (0.93), fronting (0.96), numerals (0.97), UPPER CASE and double spaces (1.00), invisible Unicode (0.95–1.00) and odd inputs (0.98) from normal inputs. It cannot see synonyms (0.54) or a verb missing its object (0.46), the same blind spots as output confidence. To predict a wrong output, output confidence is better (0.95) than input distance (0.80), and combining them does not help (0.93). Input distance is still useful where there is no correct answer at all: flagging odd inputs (0.98) gives the model a way to refuse.

4 Retraining: R1 and E1

After the stress tests, two retraining strategies were tried, both from the base model (Qwen3-0.6B-Base, fresh LoRA, glyphs starting at their Latin sounds, seed 0, 3 epochs; E1 was later repeated with 5 epochs and E1b with a

second seed), both with refusal examples (non-English input gives the output `<not-english>`) and plain-English replay passages (FineWeb-Edu) against forgetting:

- **R1, targeted:** the template data plus English rewrites and new structures aimed at the failures found, each with a *held-out* variant never used in training (e.g. passive with half of the verbs; numerals one–three trained, four–five held out).
- **E1, representative:** English generated from grammar families sampled uniformly; 20% of variants held out at random before training. **E1a** translates directly; **E1b** first writes an analysis of the English (`S=big+dog.PL V=see.PAST.NEG O=cat L=house => <glyphs>`). Same data otherwise.

4.1 R1 compared with Option 1

Source: ***results/r1_vs_option1.md****.*

Caveat

Teacher-forced scores on 100 new items per category. "[train]" / "[heldout]" refer to R1's training data. Option 1 never saw any of these variants, so its column shows the starting point.

Both: Qwen3-0.6B-Base + LoRA r=16, 3 epochs, glyphs as new tokens starting at their Latin sounds, seed 0. R1 adds varied English, new structures, refusals and plain-English replay (train/augment.py). Teacher-forced exact match. [train] = this kind of variant was in R1's training data; [heldout] = it never was (tests generalisation). Option 1 never saw any of them.

The four original test splits

split	Option 1	R1
test_iid	0.990	0.986
test_comp	0.986	0.988
test_morph	0.972	0.970
test_long	0.974	0.982

Training time: Option 1 49.6 min, R1 58.8 min.

New categories (100 items each)

category	Option 1	R1	R1 on Kaggle (check)
rewrite passive [train]	0.00	0.93	0.93
rewrite passive [heldout]	0.00	0.94	0.94
rewrite fronting [train]	0.00	0.99	0.99
rewrite fronting [heldout]	0.00	0.00	0.00
rewrite filler [train]	0.01	0.99	0.99
rewrite filler [heldout]	0.06	0.99	0.99
rewrite determiner [train]	0.16	0.98	0.98
rewrite determiner [heldout]	0.03	0.98	0.98
rewrite numeral [train]	0.00	1.00	1.00
rewrite numeral [heldout]	0.00	1.00	1.00
rewrite tense [train]	0.81	0.99	0.99
rewrite tense [heldout]	0.96	0.99	0.99
rewrite synonym [train]	0.14	0.78	0.78
rewrite synonym [heldout]	0.00	0.15	0.15
rewrite typo [train]	0.03	0.13	0.13
rewrite typo [heldout]	0.07	0.12	0.12
rewrite spacing [train]	0.05	0.99	0.99
rewrite spacing [heldout]	0.03	0.98	0.98
rewrite casing [train]	0.41	0.72	0.72
rewrite casing [heldout]	0.04	0.29	0.29
structure 2 adjectives [train]	0.09	0.96	0.96
structure 3 adjectives [heldout]	0.00	0.30	0.31
structure verb without object [train]	0.00	0.83	0.83
structure name as subject [train]	0.00	0.47	0.47
structure name as object [heldout]	0.00	0.22	0.22
structure two sentences [train]	0.00	0.90	0.90

category	Option 1	R1	R1 on Kaggle (check)
structure three sentences [heldout]	0.00	0.06	0.06
refusal [train]	0.00	1.00	1.00
refusal [heldout]	0.00	0.85	0.85

Training data used by R1

kind	examples
none	25091
passive	2156
fronting	2016
typo	2781
numeral	2607
spacing	2899
determiner	1787
filler	2950
tense	3019
synonym	1783
casing	2911
2 adjectives	1500
verb without object	1000
name as subject	1500
two sentences	1000
refusal	1500
replay	5000
names_used	1461

Inference

R1 keeps the original splits within ±1 point and fixes what it targets, generalising to held-out variants of the trained patterns (passive 0.94, numerals and determiners 0.98–1.00, refusal 0.85). It does not generalise to cases outside them: object fronting 0.00, unseen synonyms 0.15, three adjectives 0.30, three sentences 0.06, typos 0.12–0.13.

4.2 Option 1, R1, E1a and E1b on the same test items

Source: `results/e1_comparison.md`.

Caveat

Free generation, glyph part only. Rows starting 'R1' are R1's categories with R1's train/held-out labels; for E1, three adjectives and three sentences are inside its training range. E1's own held-out variants are listed above the table. E1 used 30k translation inputs and was not converged (see the per-epoch table below).

Exact match of the glyph output (free generation; for the original splits of Option 1 and R1, teacher-forced exact match, which equals greedy generation for these models). All models: Qwen3-0.6B-Base + LoRA, seed 0.

E1's randomly held-out variants (never in E1 training): case=sentence, det_sg=one, noise=drop, order=object cleft, order=object topicalised, order=subject cleft, subject=name, voice=passive. Rows starting 'R1' are R1's categories; their [train]/[heldout] labels refer to R1's training, not E1's.

group	n	Option 1	R1	E1a (plain)	E1b (analysis)
original test_iid	500	0.99	0.99	0.26	0.96
original test_comp	500	0.99	0.99	0.22	0.93
original test_morph	500	0.97	0.97	0.27	0.93
original test_long	500	0.97	0.98	0.06	0.84
in-distribution mix	100	0.09	0.24	0.30	0.76
held-out case = sentence	100	0.07	0.31	0.34	0.74
held-out det_sg = one	100	0.03	0.22	0.42	0.74
held-out noise = drop	100	0.00	0.08	0.16	0.51
held-out order = object cleft	100	0.02	0.07	0.07	0.12
held-out order = object topicalised	100	0.04	0.09	0.13	0.20
held-out order = subject cleft	100	0.00	0.00	0.02	0.07
held-out subject = name	100	0.00	0.10	0.00	0.00
held-out voice = passive	100	0.05	0.12	0.07	0.12

group	n	Option 1	R1	E1a (plain)	E1b (analysis)
held-out synonym pairs	100	0.02	0.28	0.06	0.07
extrapolation: 5-6 adjectives on a noun	100	0.08	0.11	0.25	0.58
extrapolation: 5 clauses	100	0.00	0.00	0.20	0.36
refusal [train]	100	0.00	1.00	1.00	1.00
refusal [heldout]	100	0.00	0.82	0.64	0.73
R1 rewrite passive [train]	50	0.00	0.94	0.00	0.00
R1 rewrite passive [heldout]	50	0.00	0.94	0.00	0.00
R1 rewrite fronting [train]	50	0.00	1.00	0.26	0.98
R1 rewrite fronting [heldout]	50	0.00	0.00	0.00	0.00
R1 rewrite filler [train]	50	0.02	0.96	0.40	0.68
R1 rewrite filler [heldout]	50	0.08	1.00	0.38	0.80
R1 rewrite determiner [train]	50	0.30	1.00	0.18	0.94
R1 rewrite determiner [heldout]	50	0.04	0.98	0.20	0.88
R1 rewrite numeral [train]	50	0.00	1.00	0.30	0.94
R1 rewrite numeral [heldout]	50	0.00	1.00	0.18	0.94
R1 rewrite tense [train]	50	0.76	1.00	0.38	1.00
R1 rewrite tense [heldout]	50	0.98	0.98	0.54	0.98
R1 rewrite synonym [train]	50	0.12	0.74	0.08	0.36
R1 rewrite synonym [heldout]	50	0.00	0.30	0.10	0.12
R1 rewrite typo [train]	50	0.04	0.12	0.12	0.52
R1 rewrite typo [heldout]	50	0.02	0.04	0.10	0.50
R1 rewrite spacing [train]	50	0.02	1.00	0.34	0.94
R1 rewrite spacing [heldout]	50	0.04	1.00	0.38	0.98
R1 rewrite casing [train]	50	0.56	0.78	0.24	0.94
R1 rewrite casing [heldout]	50	0.02	0.28	0.10	0.72
R1 structure 2 adjectives [train]	50	0.18	0.92	0.16	0.92
R1 structure 3 adjectives [heldout]	50	0.00	0.30	0.08	0.86
R1 structure verb without object [train]	50	0.00	0.74	0.02	0.00
R1 structure name as subject [train]	50	0.00	0.32	0.00	0.00
R1 structure name as object [heldout]	50	0.00	0.32	0.50	0.96
R1 structure two sentences [train]	50	0.00	0.88	0.10	0.88
R1 structure three sentences [heldout]	50	0.00	0.16	0.02	0.80
R1 refusal [train]	50	0.00	1.00	1.00	1.00
R1 refusal [heldout]	50	0.00	0.82	0.74	0.70

Inference

After 3 epochs on identical data, the analysis step (E1b) is far ahead of direct translation (E1a) almost everywhere (original iid 0.96 vs 0.26; three sentences 0.80 vs 0.02). Most of this gap is speed of learning, not a final difference: with 5 epochs E1a catches up on most tests (section “E1 with 5 epochs” below). E1b also counts beyond its training range (five or six adjectives 0.58) and handles held-out surface variants (sentence case 0.74, the determiner ‘one’ 0.74, dropped-letter typos 0.51). R1 is best on what it targeted (passive 0.94, synonyms it was trained on 0.74). No model handles a structure type it never saw: counting only items that really contain it, E1’s held-out passive is 0.00, clefts 0.00–0.05 and object topicalisation 0.02–0.04 (the pooled rows in the table are inflated by items whose variant fell back to normal word order; correction in results/e1_heldout_corrected.md); names as subject 0.00.

4.3 Correction: held-out structures, counting only items that contain them

Source: `results/e1_heldout_corrected.md`.

When a held-out variant cannot apply to a clause (passive needs an object and a known participle; clefts and topicalisation need the right parts), `train/represent.py` silently falls back to the canonical form, so part of each ”held-out” evaluation group does not contain the held-out structure. Items were re-classified by pattern (passive: be-form + participle + ”by”; clefts: ”it is/was/will be ... that”; topicalised: a fronted noun phrase followed by a comma). Exact match of the glyph output, from the per-item results of the E1 runs.

held-out group	items WITH the structure	E1a	E1b	items that fell back	E1a	E1b
voice = passive	62	0.00	0.00	38	0.18	0.32
order = subject cleft	76	0.01	0.05	24	0.04	0.12
order = object cleft	66	0.00	0.02	34	0.21	0.32
order = object topicalised	81	0.02	0.04	19	0.58	0.89

Conclusion (stronger than before): structure types absent from training are essentially never handled (0.00-0.05).

The pooled figures in e1_comparison.md (0.07-0.20) are inflated by the fallback items. Fixed for future runs: `represent.eval_sets(..., strict=True)` keeps only items in which the forced variant was actually used.

Inference

Unseen structure types are essentially never handled by either E1 model (0.00–0.05 on the items that really contain them). Items whose variant fell back to the ordinary form are translated much better (up to 0.89), which is what inflated the pooled group scores. Future runs build these groups only from items that really contain the variant.

4.4 Forgetting, adapter scaling and per-epoch checkpoints

Source: `results/followups_e1.md`.

Caveat

WikiText-2: 40 chunks of 256 tokens (never trained on). Scaling and per-epoch cells use subsets (50 and 30 items per group; 100 per original split), about ±7–9 points.

5. English forgetting

WikiText-2 test: 40 chunks x 256 tokens (never trained on). Everyday: the 30 sentences of test 21. 'Masked' = glyph tokens removed from the prediction. Leak = plain-English continuations that contain glyphs.

model	WikiText-2 ppl	masked	everyday ppl	masked	probability on glyphs	leak
Base Qwen3-0.6B-Base	22.9	22.9	20.3	20.3	0.000	0%
Option 1	4207.6	521.9	372.0	246.7	0.195	47%
R1	35.6	34.9	22.7	22.7	0.000	0%
E1a (plain)	34.3	33.6	23.5	23.5	0.000	0%
E1b (analysis)	35.5	34.9	23.0	23.0	0.000	0%

6. E1b with the LoRA adapter scaled at inference

measure	strength 0.25	strength 0.5	strength 0.75	strength 1.0
WikiText-2 ppl (20 chunks)	24.6	26.2	29.6	36.0
everyday ppl	18.9	19.0	20.4	23.0
in-distribution mix	0.00	0.00	0.08	0.70
held-out voice = passive	0.00	0.00	0.04	0.14
held-out synonym pairs	0.00	0.00	0.02	0.06
R1 rewrite synonym [train]	0.00	0.00	0.00	0.36
R1 rewrite typo [train]	0.00	0.00	0.02	0.52
original test_iid (first 200)	0.00	0.00	0.02	0.94

7. Per-epoch checkpoints (exact match of the glyph output)

Subsets for laptop time: first 100 items of each original split (label says 200: name kept for the lookup), first 30 of every other group, identical for every checkpoint (about +/-9 points per cell).

group	R1 ep1	R1 ep2	R1 ep3	E1a (plain) ep1	E1a (plain) ep2	E1a (plain) ep3	E1b (analysis) ep1	E1b (analysis) ep2	E1b (analysis) ep3
in-distribution mix	0.03	0.37	0.33	0.07	0.20	0.33	0.10	0.30	0.77
held-out case = sentence	0.13	0.20	0.30	0.07	0.27	0.33	0.07	0.30	0.87
held-out noise = drop	0.00	0.07	0.10	0.00	0.00	0.00	0.00	0.07	0.43
held-out order = object topicalised	0.00	0.00	0.00	0.00	0.07	0.07	0.07	0.07	0.13
held-out voice = passive	0.00	0.13	0.13	0.03	0.03	0.03	0.07	0.07	0.07
held-out synonym pairs	0.03	0.23	0.37	0.00	0.03	0.07	0.03	0.00	0.07
extrapolation: 5-6 adjectives on a noun	0.13	0.17	0.17	0.13	0.20	0.23	0.17	0.27	0.60
extrapolation: 5 clauses	0.00	0.00	0.00	0.00	0.10	0.30	0.00	0.20	0.30
refusal [heldout]	0.93	0.80	0.80	0.90	0.83	0.60	0.83	0.83	0.77
R1 rewrite synonym [train]	0.07	0.67	0.80	0.00	0.03	0.07	0.03	0.07	0.40
R1 rewrite typo [train]	0.00	0.10	0.13	0.03	0.07	0.13	0.03	0.07	0.60
R1 structure 3 adjectives [heldout]	0.00	0.23	0.37	0.00	0.00	0.13	0.00	0.17	0.80
R1 structure three sentences [heldout]	0.00	0.10	0.13	0.00	0.00	0.00	0.00	0.00	0.87

group	R1 ep1	R1 ep2	R1 ep3	E1a (plain) ep1	E1a (plain) ep2	E1a (plain) ep3	E1b (analysis) ep1	E1b (analysis) ep2	E1b (analysis) ep3
original test_iid (first 200)	0.07	0.92	1.00	0.03	0.12	0.33	0.06	0.39	0.94
original test_comp (first 200)	0.00	0.88	0.98	0.00	0.01	0.27	0.00	0.21	0.92
original test_morph (first 200)	0.06	0.85	0.99	0.04	0.10	0.26	0.06	0.25	0.92
original test_long (first 200)	0.00	0.78	1.00	0.00	0.00	0.05	0.00	0.06	0.90

Inference

Replay removes forgetting (everyday-English perplexity 22.7–23.5 vs 20.3 base and 372 for Option 1; no glyph leakage). Scaling the adapter down is not a free fix: translation appears only at full strength. R1 has learned most of what it learns by epoch 2; E1a and E1b are still improving steeply at epoch 3 (E1b original iid 0.06 / 0.39 / 0.94), so the E1 runs need more epochs (done: next section). Refusal is learned first (0.83–0.93 after epoch 1).

4.5 E1 with 5 epochs, and a second E1b seed

Source: `results/e1_5ep.md`.

Caveat

5-epoch runs on a RunPod V100 (fp16, torch 2.6, transformers 5.16.1), otherwise the same settings as the 3-epoch Kaggle runs. Held-out structure rows are strict for the 5-epoch runs and left blank for the 3-epoch runs (their pooled values are not comparable). Seed 1 is a different random language, so its test items differ. 100 items per E1 group: about ±8–10 points.

Free generation, exact match of the glyph output. 5-epoch runs: RunPod V100 (fp16, torch 2.6, transformers 5.16.1), otherwise identical settings to the 3-epoch Kaggle runs. Held-out voice/order groups of the 5-epoch runs are strict (only items that contain the variant); the 3-epoch cells for those rows are left blank (see e1_heldout_corrected.md). Seed 1 = a new random language instance (different test items). n = 500 per original split, 100 per E1 group, 50 per R1 group.

group	n	E1a 3ep	E1b 3ep	E1a 5ep	E1b 5ep	E1b 5ep seed 1
original test_iid	500	0.26	0.96	0.98	0.99	0.97
original test_comp	500	0.22	0.93	0.97	0.97	0.97
original test_morph	500	0.27	0.93	0.96	0.97	0.96
original test_long	500	0.06	0.84	0.85	0.87	0.91
in-distribution mix	100	0.30	0.76	0.76	0.80	0.78
held-out case = sentence	100	0.34	0.74	0.76	0.80	0.86
held-out det_sg = one	100	0.42	0.74	0.80	0.83	0.85
held-out noise = drop	100	0.16	0.51	0.42	0.54	0.57
held-out order = object cleft	100			0.01	0.02	0.04
held-out order = object topicalised	100			0.00	0.00	0.01
held-out order = subject cleft	100			0.02	0.09	0.02
held-out subject = name	100	0.00	0.00	0.02	0.00	0.00
held-out voice = passive	100			0.00	0.01	0.02
held-out synonym pairs	100	0.06	0.07	0.13	0.10	0.17
extrapolation: 5-6 adjectives on a noun	100	0.25	0.58	0.53	0.71	0.75
extrapolation: 5 clauses	100	0.20	0.36	0.48	0.48	0.58
refusal [train]	100	1.00	1.00	1.00	1.00	1.00
refusal [heldout]	100	0.64	0.73	0.76	0.79	0.83
R1 structure 2 adjectives [train]	50	0.16	0.92	0.98	1.00	1.00
R1 structure 3 adjectives [heldout]	50	0.08	0.86	0.94	0.98	0.98
R1 structure verb without object [train]	50	0.02	0.00	0.02	0.02	0.00
R1 structure name as subject [train]	50	0.00	0.00	0.00	0.00	0.00
R1 structure name as object [heldout]	50	0.50	0.96	1.00	0.98	0.92
R1 structure two sentences [train]	50	0.10	0.88	0.94	0.96	0.96
R1 structure three sentences [heldout]	50	0.02	0.80	0.98	0.86	0.94
R1 refusal [train]	50	1.00	1.00	1.00	1.00	1.00
R1 refusal [heldout]	50	0.74	0.70	0.78	0.78	0.76

run	training minutes	target tokens/s
E1a 3ep	54	1743
E1b 3ep	74	1962
E1a 5ep	76	2089
E1b 5ep	88	2737
E1b 5ep seed 1	88	2788

Inference

The analysis step was useful and made the model converge faster. With the analysis, 3 epochs were enough (E1b original splits 0.96 / 0.93 / 0.93 / 0.84); without it, the same data and schedule left E1a at 0.26 / 0.22 / 0.27 / 0.06, and E1a needed 5 epochs to reach that level (0.98 / 0.97 / 0.96 / 0.85). Counting compute rather than epochs, the saving is smaller but still real: an E1b epoch costs about 1.4× an E1a epoch (longer outputs), so 3 E1b epochs (74 T4-minutes) against 5 E1a epochs (about 90 T4-minutes) is about 1.2× less compute. At 5 epochs the two versions are close on most tests, and E1b's two seeds agree within about 5 points on most rows, so the results hold on a second random language. E1b keeps an edge on both seeds on the hardest generalisation: five or six adjectives 0.71 / 0.75 vs 0.53, typos 0.54 / 0.57 vs 0.42. Longer training does not help unseen structure types: passive, clefts, object topicalisation and names as subject stay at 0.00–0.09 for every model. With representative data and enough training, the model generalises to new surface forms and to more of the same structure, but not to a structure type it has never seen.

4.6 Where E1b goes wrong: understanding vs translation

Source: `results/e1b_error_breakdown.md`.

Caveat

For each item: is the analysis E1b writes exactly right, and is the glyph output right? Held-out structure rows here use the pooled (uncorrected) groups.

For every item: was the analysis the model wrote exactly right, and was the glyph output right? Source: eval_gen per-item results of e1b-hard-rep-cot-50k.

group	n	both right	analysis right, glyphs WRONG (translation error)	analysis WRONG, glyphs right	both wrong (understanding error)
original test_iid	500	96%	3%	0%	1%
original test_comp	500	93%	5%	0%	2%
original test_morph	500	93%	5%	0%	1%
original test_long	500	84%	14%	0%	3%
in-distribution mix	100	76%	6%	0%	18%
held-out case = sentence	100	74%	6%	0%	20%
held-out det_sg = one	100	72%	3%	2%	23%
held-out noise = drop	100	51%	7%	0%	42%
held-out order = object cleft	100	12%	1%	0%	87%
held-out order = object topicalised	100	20%	1%	0%	79%
held-out order = subject cleft	100	7%	0%	0%	93%
held-out subject = name	100	0%	3%	0%	97%
held-out voice = passive	100	12%	2%	0%	86%
held-out synonym pairs	100	4%	0%	3%	93%
extrapolation: 5-6 adjectives on a noun	100	58%	19%	0%	23%
extrapolation: 5 clauses	100	36%	32%	0%	32%
all groups pooled	3200	70%	7%	0%	23%

Inference

E1b's failures on unseen structures are understanding errors: the analysis is already wrong in 79–97% of held-out passive, cleft, topicalised, name-subject and synonym items, and the glyphs then follow the wrong analysis. When the analysis is right, the glyphs are almost always right (translation errors 0–7% on most groups). The exception is long output: on five clauses 32% of items have a correct analysis but wrong glyphs, and on the original long split 14%. Pooled over all groups: 70% both right, 7% translation errors, 23% understanding errors. Better English understanding, not better glyph writing, is what the remaining errors need.

4.7 Inference-time fixes without retraining: sentence splitting and lower-casing

Source: `results/local_fixes.md`.

Caveat

50 items per group, free generation. Splitting breaks the input at '.', ';', '!' and ', and', translates each part and joins the outputs. Lower-casing changes the input only.

50 items per group from E1's evaluation groups; exact match of the glyph output. Cells: as given → with the fix. Split rule: break at '.', ';', '!' and ', and '; translate each part; join. Lower-casing also lower-cases names (none in these groups' casing rows except the in-distribution mix).

fix	group	Option 1	R1	E1a (plain)	E1b (analysis)
split	R1 structure two sentences [train]	0.00 → 1.00	0.88 → 1.00	0.10 → 0.06	0.88 → 0.88
split	R1 structure three sentences [heldout]	0.00 → 0.98	0.16 → 1.00	0.02 → 0.06	0.82 → 0.84
split	extrapolation: 5 clauses	0.00 → 0.00	0.00 → 0.02	0.16 → 0.22	0.30 → 0.54
split	in-distribution mix	0.10 → 0.12	0.34 → 0.34	0.28 → 0.28	0.70 → 0.74
lowercase	R1 rewrite casing [train]	0.56 → 1.00	0.78 → 0.96	0.24 → 0.26	0.94 → 0.96
lowercase	R1 rewrite casing [heldout]	0.02 → 1.00	0.28 → 0.98	0.10 → 0.18	0.72 → 0.98
lowercase	held-out case = sentence	0.06 → 0.06	0.32 → 0.30	0.34 → 0.36	0.82 → 0.78
lowercase	in-distribution mix	0.10 → 0.12	0.34 → 0.32	0.28 → 0.24	0.70 → 0.68

Inference

Both fixes repair a failure completely when it is the only thing wrong with the input. Splitting lifts Option 1 from 0.00 to 1.00 on two sentences and to 0.98 on three (never seen), and R1 from 0.16 to 1.00 on three; lower-casing lifts the casing variants of R1's categories to 0.96–1.00 for Option 1 and R1. They do not help when the input also differs in other ways: the E1 groups (5 clauses, sentence case, the in-distribution mix) are E1-style English with several kinds of variation at once, and there Option 1 and R1 stay at 0.00–0.34 with or without the fix. E1a gains little anywhere (these are the 3-epoch E1 models; E1a was undertrained). E1b already handles multiple sentences without splitting (0.88 / 0.82); splitting helps it on five clauses (0.30 to 0.54), which matches the error breakdown above (long outputs were its main translation error), and lower-casing lifts its unseen casing variants from 0.72 to 0.98. Cheap input normalisation is worth applying in front of any of these models, but it cannot replace representative training data.

5 Overall inference

1. The research question

Token count does not predict accuracy; the meaning a token starts with does. Tamil script costs 2.2–3.4× the tokens of Latin and learns as well or faster; splitting Latin into single characters (more tokens) helps the hard language; invented glyphs fail as 3 byte-tokens *and* as single new tokens with an uninformative start. Starting each glyph at its Latin sound (Option 1) gives 0.990 / 0.986 / 0.972 / 0.974 on the four splits, above both real scripts, and the shuffled-sound Control shows that the *correct* sounds are what produce generalisation to new structures (long 0.974 vs 0.650).

2. What Option 1 really learned

Within the kind of English it was trained on, Option 1 learned the invented grammar as rules: held-out ending combinations 95.0% (test 3), the never-seen sentence shape 97.5% (test 20), never-seen adjective+noun pairs 100% (test 20), one-feature changes 99.2–100% with no side effects (test 14), no agreement attraction (test 13), the correct form preferred over every one-error version 100% (test 15), and rare words learned from 1–15 sightings (99.8%, test 2). Its glyph embeddings keep their sound layout (test 1), its outputs are well formed (tests 11, 16), and it is well calibrated in distribution (test 29).

3. Where it fails, and one mechanism behind most failures

It fails whenever the English leaves the training template: passive 0%, fronting 0%, fillers 0%, numerals 0%, determiners 6%, synonyms 12–19%, typos 3–10%, UPPER CASE 4–18%, invisible Unicode 1–34% (tests 4, 8, 17, 19, 22, 24, 27, 30; the range covers every test that measured it); and on structures absent from training: two adjectives on a noun, two sentences, a verb without its object, names (tests 5, 6, 7, 10). Three follow-up tests show how: it translates *word by word from the English tokens* (an error stays on the changed word in 95–97% of cases, test 22), and it fills a *fixed positional template*: a passive sentence is read as "first noun = doer" (swapped translation scored higher in 100% of cases, test 23), and two adjectives are forced into a one-adjective slot (test 25). An "unseen local structure" flag catches 81.7% of the failures (test 27); what it misses are inputs where something is missing rather than new (a verb without its object) or changed inside the whitespace. **All four models fail in the same ways**, so these are consequences of the narrow, single-template training English, not of the glyph start.

4. Can the failures be caught or fixed without retraining?

- **Not by decoding:** beam search does not help and almost all errors are model errors (test 17).
- **Partly by confidence:** the model's confidence in its own output separates right from wrong (0.93–0.95, also within input types, tests 17, 24, 28), but it is over-confident off-template and blind to synonym errors (0.63; tests 24, 29), so at 99% accuracy only 3.0% of a mixed off-template pool could be answered (test 29).
- **Refusal:** the model never refuses (test 16), but input distance flags odd inputs with 0.98 AUROC (test 28), a usable switch.
- **Input clean-up:** NFKC + removing invisible characters fully repairs zero-width spaces, soft hyphens, no-break spaces and full-width letters (test 30). Look-alike letters need a confusables map. (Lower-casing as a fix for UPPER CASE is plausible but was not tested.)

5. Side effect: forgetting English

Fine-tuning damaged plain English (perplexity 20.3 to 372 for Option 1, to millions for Options 2 and 3a; test 21). The damage is in the LoRA adapter, not the glyph rows (test 26): adapter off restores 20.5–23.2, and removing glyphs from the prediction removes only the smaller part of the damage. The adapters of the copy-start models damaged English 100–500× more; why is not established (one hypothesis: their adapters had to do more of the work of identifying glyphs).

6. Retraining with targeted data (R1): fixes what it targets, and close variants

R1 (Option 1 + varied English rewrites + new structures + refusals + plain-English replay, each with a held-out variant) keeps the original splits (0.986 / 0.988 / 0.970 / 0.982) and generalises *within* the patterns it was given: passive with verbs never seen in passive 0.94, held-out fillers / determiners / numerals / spacing 0.98–1.00, refusal of unseen kinds of non-English 0.85. It does not reach neighbouring cases that were not given: object fronting 0.00 (location fronting 0.99), unseen synonyms 0.15, three adjectives 0.30, three sentences 0.06; typos stay at 0.13.

7. Representative English (E1) and an analysis step

E1 replaces the single template with English generated from grammar families (voice, word order, tense expression, negation form, determiners, adjectives, location preposition, clause joining, orthography, lexical choice), sampled uniformly, with 20% of variants held out at random before training. Both versions were trained for 3 and for 5 epochs; E1b was also repeated on a second random language (seed 1). **The analysis step was useful and made the model converge faster:** after 3 epochs the version that first writes an analysis of the English (E1b) already scored 0.96 / 0.93 / 0.93 / 0.84 on the original splits, while the plain version (E1a) on the same data and schedule was at 0.26 / 0.22 / 0.27 / 0.06; E1a needed 5 epochs to reach the same level (0.98 / 0.97 / 0.96 / 0.85). At 5 epochs the gap on most tests closes (E1b 0.99 / 0.97 / 0.97 / 0.87, seed 1 0.97 / 0.97 / 0.96 / 0.91), but E1b keeps an edge on both seeds where generalisation is hardest: five or six adjectives (never trained) 0.71 / 0.75 vs 0.53, dropped-letter typos 0.54 / 0.57 vs 0.42. Held-out surface variants reach 0.76–0.86 for both. **Structure types that never appear in training still fail for every model, at every length of training and with or without the analysis** (strict groups, only items that contain the structure: passive 0.00–0.02, clefts 0.01–0.09, object topicalisation 0.00–0.01, names as subject 0.00–0.02), and so do unseen synonym pairs (0.10–0.17).

8. Forgetting is fixed by replay

With about 10% plain-English passages mixed into training, English perplexity is back near the base model (everyday sentences 22.7–23.5 vs 20.3; WikiText-2 34.3–35.6 vs 22.9) instead of 372 / 4,208 for Option 1, and glyphs no longer leak into English (0% vs 47%). Scaling the adapter down at inference is not an alternative: E1b translates almost nothing below full strength (0.02 on the original iid split at strength 0.75).

9. Summary

1. What makes an unseen script learnable is the meaning its new tokens start with, not their number.
2. A fine-tuned model generalises within what its data covers: new words, new combinations, more of the same, new surface forms. Structure types absent from training are not reached, whether the data is targeted (R1) or broad (E1). The training distribution should therefore cover every structure type the language should handle, chosen in advance from a grammar, not from test failures.
3. An explicit analysis step was useful: it made the model converge faster (3 epochs instead of 5 for the same scores) and kept a consistent edge on the hardest generalisation (counting beyond the trained range, typos) on two random languages. It does not help with structure types absent from training.
4. Plain-English replay removes forgetting; refusal of non-English generalises to unseen kinds (0.64–0.85).

6 Caveats and limits

- **One seed per 50k model.** Only the 20k grid has three seeds (sd 0.010–0.023). Differences of a few points between the 50k models may be noise; the large gaps (e.g. long 0.974 vs 0.650) are not.
- **One invented language** (seed 0) for all stress tests; another random language with the same grammar might differ in detail.
- **Small cells:** stress-test rows have 60–300 items (test 15's irregular-verb row has 8), so a difference of a few points within a row is not meaningful.
- **Assumed gold** for structures never in training (two adjectives); test 25 shows the conclusion does not depend on it.
- **Forgetting** is measured on 30 everyday sentences by perplexity, which is very sensitive to probability on glyph tokens.
- **Synthetic English:** all English comes from the generator's templates; real English would be more varied and would likely expose more failures.
- **Untested hypotheses** are marked as such in the text (why copy-start adapters forget more; lower-casing as a fix).
- Test 12 is invalid and is not used anywhere in the conclusions.
- **R1 and E1a are single runs (seed 0); E1b (5 epochs) has two seeds.** E1 used 30k translation inputs (Option 1 and R1: 50k); E1's held-out list is one random draw. The E1a/E1b edge at 5 epochs (about 20 points on two groups of 100) compares two E1b seeds against one E1a seed.
- **The 5-epoch runs ran on a RunPod V100** (torch 2.6) instead of a Kaggle T4 (torch 2.11); all other settings are identical. An E1b epoch costs more compute than an E1a epoch (longer outputs: 74 vs 54 training minutes for 3 epochs on the T4), so the faster convergence is about 1.2× in compute rather than 5/3 in epochs.
- **Per-epoch cells use subsets** (100 items per original split, 30 per other group: about ±9 points).
- **A sealed set of independently written English** has not been scored yet; all English so far is generated.

A Where every number comes from

- Main tables: `results/curve_summary.md` (`analysis/curve_table.py`) and `results/final_20k.md` (`analysis/final_20k.py`), built from the per-run JSON files in `results/kaggle/<run>/results/`. Figures: `analysis/plots.py`.
- Stress tests: `analysis/stress_tests.py` (1–6), `stress_tests2.py` (7–12), `stress_tests3.py` (13–21), `stress_tests4.py` (22–26), `stress_tests5.py` (27–30); each writes `results/stress/test<N>_*.md`. Saved models: `results/kaggle/s1-hard-latininit-50k` (Option 1), `s2-hard-translit-50k` (Option 2), `s3-hard-plaintext-50k` (Option 3a), `s4-hard-shuffledinit-50k` (Control); the reruns reproduced the original runs' scores exactly.
- Split examples in Section 1 are produced by `conlang.dataset.build` at report time.
- Retraining: data from `train/augment.py` (R1) and `train/represent.py` (E1); runs `results/kaggle/r1-hard-aug-50k`, `e1a-hard-rep-50k`, `e1b-hard-rep-cot-50k` (each with `model/`, `model_epoch1/`, `model_epoch2/`). Comparisons: `analysis/compare_r1.py`, `analysis/compare_e1.py`, `analysis/followups_e1.py`, `analysis/compare_5ep.py` (5-epoch runs `e1a-hard-rep-5ep`,

`e1b-hard-rep-cot-5ep`, `e1b-hard-rep-cot-5ep-seed1`, trained on RunPod with `kaggle_tools/runpod_e1.sh`), `analysis/local_fixes.py`. Literature behind the E1 design: `results/generalisation_research.md`.

- This report: `docs/results_report/build_report.py` (tables) and `inferences.py` (text).